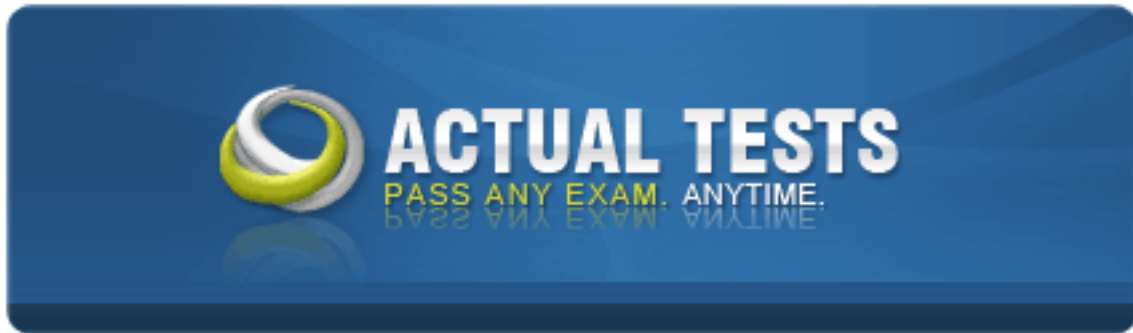


Cisco 642-637



**Securing Networks with Cisco Routers and Switches
(SECURE) v1.0**

Version: 8.0

QUESTION NO: 1

Refer to the exhibit. Given the partial output of the debug command, what can be determined?

```
Router# debug crypto isakmp
*ISAKMP (1009): received packet from 192.168.2.2 dport 500 sport 500 Global (I)
MM_KEY_EXCH
ISAKMP:(1009): processing ID payload. message ID = 0
ISAKMP (1009): ID payload
    next-payload : 8
    type         : 1
    address      : 192.168.2.2
    protocol     : 17
    port         : 500
    length       : 12
ISAKMP:(0):: peer matches *none* of the profiles
ISAKMP:(1009): processing HASH payload. message ID = 0
ISAKMP:(1009): SA authentication status:      authenticated
ISAKMP:(1009): SA has been authenticated with 192.168.2.2
```

- A. There is no ID payload in the packet, as indicated by the message ID = 0.
- B. The peer has not matched any offered profiles.
- C. This is an IKE quick mode negotiation.
- D. This is normal output of a successful Phase 1 IKE exchange.

Answer: B

Explanation:

Although the authentication of IKE phase 1 is authenticated, the exhibit question says "Given the partial output of the "debug command", what can be determined? 2 is best for the peer has not matched any offered profiles.

QUESTION NO: 2 DRAG DROP

Drag the items on the left to the boxes on the right that identify important information you should collect prior to deploying 802.1X authentication in a Cisco IBNS environment. Not all items will be used.

- existing list of LAN switches
- existing user credentials
- existing addressing scheme
- existing automated software deployment mechanisms
- existing transport protocols used in the environment
- trustworthiness of the existing transport network

Answer:

Drag the items on the left to the boxes on the right that identify important information you should collect prior to deploying 802.1X authentication in a Cisco IBNS environment. Not all items will be used.

existing list of LAN switches	existing list of LAN switches
existing user credentials	existing user credentials
existing addressing scheme	existing automated software deployment mechanisms
existing automated software deployment mechanisms	trustworthiness of the existing transport network
existing transport protocols used in the environment	
trustworthiness of the existing transport network	

Explanation:

Drag the items on the left to the boxes on the right that identify important information you should collect prior to deploying 802.1X authentication in a Cisco IBNS environment. Not all items will be used.

	existing list of LAN switches
	existing user credentials
existing addressing scheme	existing automated software deployment mechanisms
	trustworthiness of the existing transport network
existing transport protocols used in the environment	

Page 113 of the CCNP Secure guide Gathering Input Parameters

Because 802.1X authentication requires several technologies to work together, up-front planning helps ensure the success of the deployment.

Part of this planning involves gathering important input information:

QUESTION NO: 3

Refer to the exhibit.

```
webvpn context MY-WEBVPN
policy group GROUP-POLICY
functions svc-enabled
svc keep-client-installed
svc address-pool MY-POOL
svc default-domain cisco.com
svc dns-server primary 10.10.1.1
svc split dns domain.com
svc split include 10.0.0.0 255.0.0.0
filter tunnel FILTER-ACL
```

Which two Cisco IOS WebVPN features are enabled with the partial configuration shown?
(Choose two.)

- A. The end-user Cisco AnyConnect VPN software will remain installed on the end system.
- B. If the Cisco AnyConnect VPN software fails to install on the end-user PC, the end user cannot use other modes.
- C. Client based full tunnel access has been enabled.
- D. Traffic destined to the 10.0.0.0/8 network will not be tunneled and will be allowed access via a split tunnel.
- E. Clients will be assigned IP addresses in the 10.10.0.0/16 range.

Answer: A,C

Explanation:

QUESTION NO: 4

Which two of these are benefits of implementing a zone-based policy firewall in transparent mode?
(Choose two.)

- A. Less firewall management is needed.
- B. It can be easily introduced into an existing network.
- C. IP readdressing is unnecessary.
- D. It adds the ability to statefully inspect non-IP traffic.
- E. It has less impact on data flows.

Answer: B,C

Explanation:

QUESTION NO: 5

When configuring a zone-based policy firewall, what will be the resulting action if you do not specify any zone pairs for a possible pair of zones?

- A. All sessions will pass through the zone without being inspected.
- B. All sessions will be denied between these two zones by default.
- C. All sessions will have to pass through the router "self zone" for inspection before being allowed to pass to the destination zone.
- D. This configuration statelessly allows packets to be delivered to the destination zone.

Answer: B

Explanation:

Zone Pair Configuration

The configuration of the zone pair is important because its configuration dictates the direction in which traffic is allowed to flow. As stated previously, a zone pair is unidirectional and is the part of the configuration that controls traffic between zones; this is referred to as interzone. If no zone pair is defined, traffic will not flow between zones

QUESTION NO: 6

Refer to the exhibit. What can be determined from the output of this show command?

```
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
192.168.1.1  192.168.2.1  QM_IDLE       1002 ACTIVE
```

- A. The IPsec connection is in an idle state.
- B. The IKE association is in the process of being set up.

- C. The IKE status is authenticated.
- D. The ISAKMP state is waiting for quick mode status to authenticate before IPsec parameters are passed between peers
- E. IKE Quick Mode is in the idle state, indicating a problem with IKE phase 1.

Answer: C

Explanation:

Verify Local IKE Sessions

Use the show crypto isakmp sa command to display the current IKE Security Associations (SA) on the local router. The QM_IDLE status indicates successful establishment of the IKE SA, meaning that the ISAKMP process is idle after having successfully negotiated and established SAs.

Example 15-5 shows the output of the show crypto isakmp sa command.

QUESTION NO: 7 DRAG DROP

Drag the cryptographic CLI commands on the left to their definitions on the right. Not all items will be used.

show crypto map	delete IPsec security association
show crypto isakmp sa	verify cryptographic configurations and show SA lifetimes
clear crypto sa	verify the IPsec protection policy settings
show crypto isakmp policy	verify current IPsec settings in use by the SAs
clear crypto isakmp	clear active IKE connections
show crypto ipsec sa	
show crypto ipsec transform-set	

Answer:

Drag the cryptographic CLI commands on the left to their definitions on the right. Not all items will be used.

show crypto map

show crypto isakmp sa

clear crypto sa

show crypto isakmp policy

clear crypto isakmp

show crypto ipsec sa

show crypto ipsec transform-set

clear crypto sa

show crypto isakmp policy

show crypto ipsec transform-set

show crypto ipsec sa

clear crypto isakmp

Explanation:

Drag the cryptographic CLI commands on the left to their definitions on the right. Not all items will be used.

show crypto map

show crypto isakmp sa

clear crypto sa

show crypto isakmp policy

show crypto ipsec transform-set

show crypto ipsec sa

clear crypto isakmp

Verify cryptographic configs

router# show crypto isakmp policy

rotection suite priority 15

ncryption algorithm: DES - Data Encryption Standard (56 bit keys)

ash algorithm: Message Digest 5

uthentication method: Rivest-Shamir-Adleman Signature

iffie-Hellman Group: #2 (1024 bit)

ifetime: 5000 seconds, no volume limit

rotection suite priority 20

ncryption algorithm: DES - Data Encryption Standard (56 bit keys)

ash algorithm: Secure Hash Standard authentication method: preshared Ke

QUESTION NO: 8

You are running Cisco IOS IPS software on your edge router. A new threat has become an issue. The Cisco IOS IPS software has a signature that can address the new threat, but you previously retired the signature. You decide to unretire that signature to regain the desired protection level. How should you act on your decision?

- A.** Retired signatures are not present in the routers memory. You will need to download a new signature package to regain the retired signature.
- B.** You should re-enable the signature and start inspecting traffic for signs of the new threat.
- C.** Unretiring a signature will cause the router to recompile the signature database, which can temporarily affect performance.
- D.** You cannot unretire a signature. To avoid a disruption in traffic flow, it's best to create a custom signature until you can download a new signature package and reload the router.

Answer: C

Explanation:

Some signatures can be retired. This signature is not present in the router's memory. Unretiring a retired signature requires that the router recompile the signature database.

This can temporarily affect performance and take a long time with a large signature database.

QUESTION NO: 9

Which statement best describes inside policy based NAT?

- A.** Policy NAT rules are those that determine which addresses need to be translated per the enterprise security policy
- B.** Policy NAT consists of policy rules based on outside sources attempting to communicate with inside endpoints.
- C.** These rules use source addresses as the decision for translation policies.
- D.** These rules are sensitive to all communicating endpoints.

Answer: A

Explanation:

The original dump had this option:

A) Policy NAT rules are those that determine which addresses need to be translated per the enterprise security policy

The newer dump did not so no sure the answer is still A)

http://www.cisco.com/en/US/docs/security/asa/asa82/configuration/guide/nat_overview.html#wp1088419

QUESTION NO: 10

Refer to the exhibit. What can be determined about the IPS category configuration shown?

```
ip ips signature-category
category all
  enabled false
  retired true
category os ios
  enabled true
  retired false
event-action produce-alert reset-tcp-connection
```

- A. All categories are disabled.
- B. All categories are retired.
- C. After all other categories were disabled, a custom category named "os ios" was created
- D. Only attacks on the Cisco IOS system result in preventative actions.

Answer: D

Explanation:

This configuration task is completed by entering the signature category configuration mode using the `ip ips signature-category` command. See Example 13-3 for the relevant configuration. First, retire and disable all signatures because only the desired signatures will be enabled. This is achieved using the `category all` command. Then, use the `retired true` and `enabled false` commands to disable and retire all signatures by default. Next, enable all signatures that are designed to prevent attacks against Cisco IOS Software devices and assign a preventative action to them. Enter the category that comprises these signatures using the `category os ios` command and enable them by using the `retired false` and `enabled true` commands. Use the `event-action produce-alert deny-packet-inline` command to enable these signatures to generate an alert and drop the offending packets when they trigger.

QUESTION NO: 11

When Cisco IOS IPS is configured to use SDEE for event notification, how are events managed?

- A.** They are stored in the router's event store and will allow authenticated remote systems to pull events from the event store.
- B.** All events are immediately sent to the remote SDEE server.
- C.** Events are sent via syslog over a secure SSUTLS communications channel.
- D.** When the event store reaches its maximum configured number of event notifications, the stored events are sent via SDEE to a remote authenticated server and a new event store is created.

Answer: A

Explanation:

SDEE uses a pull communication model for event messages. This allows management consoles to pull alerts from the Cisco IPS sensors over an HTTPS connection.

When Cisco SDEE notification is enabled, by default, 200 events can be stored in the local event store. This number can be increased to hold a maximum of 1000. All stored events are lost if SDEE notifications are disabled, and a new local event store is allocated when the notification feature is enabled again.

QUESTION NO: 12

Which two of these will match a regular expression with the following configuration parameters?
[a-zA-Z][0-9][a-z] (Choose two.)

- A.** Q3h
- B.** B4Mn
- C.** aaB132AA
- D.** c7lm
- E.** BBpjrIT

Answer: A,D

Explanation:

QUESTION NO: 13

Which of these is a configurable Cisco IOS feature that triggers notifications if an attack attempts to exhaust critical router resources and if preventative controls have been bypassed or are not working correctly?

- A. Control Plane Protection
- B. Management Plane Protection
- C. CPU and memory thresholding
- D. SNMPv3

Answer: C

Explanation:

CPU and Memory Thresholding

One of the ways to monitor whether an attack is occurring on a device is through the simple monitoring of device resources, including CPU and memory utilization. This is done by configuring the use of CPU or memory threshold monitoring. Both of these features can be combined with a remote management server to notify an organization when the CPU and memory conditions on a device become critical.

“With CPU Thresholding Notification, users can configure CPU utilization thresholds, which trigger a notification when exceeded. Cisco IOS Software supports two CPU utilization thresholds:”

http://www.cisco.com/en/US/products/ps6642/products_data_sheet09186a00801f98de.html

QUESTION NO: 14

Which Cisco IOS IPS feature allows to you remove one or more actions from all active signatures based on the attacker and/or target address criteria, as well as the event risk rating criteria?

- A. signature event action filters
- B. signature event action overrides
- C. signature attack severity rating
- D. signature event risk rating

Answer: A

Explanation:

QUESTION NO: 15

You are troubleshooting reported connectivity issues from remote users who are accessing corporate headquarters via an IPsec VPN connection. What should be your first step in troubleshooting these issues?

- A. issue a show crypto isakmp policy command to verify matching policies of the tunnel endpoints
- B. ping the tunnel endpoint
- C. run a traceroute to verify the tunnel path
- D. debug the connection process and look for any error messages in tunnel establishment

Answer: B

Explanation:

Page 398 - Very Important - several Questions from this
Troubleshooting Flow

Follow these steps to proceed through the recommended flow for troubleshooting IKE peering:

Step 1. Verify peer reachability using the ping and traceroute commands with the tunnel source and destination IP addresses on both peers. If connectivity is verified, proceed to Step 2; otherwise, check the path between the two peers for routing or access (firewall or access list) issues.

Step 2. Verify the IKE policy on both peers using the show crypto isakmp policy command. Debug messages revealed by the debug crypto isakmp command will also point out IKE policy mismatches.

Step 3. Verify IKE peer authentication. The debug crypto isakmp command will display unsuccessful authentication. Step 4. Upon successful completion of Steps 1–3, the IKE SA should be establishing. This can be verified with the show crypto isakmp sa command and looking for a state of QM_IDLE.

QUESTION NO: 16

Which of these is correct regarding the configuration of virtual-access interfaces?

- A. They cannot be saved to the startup configuration.
- B. You must use static routes inside the tunnels.
- C. DVTI interfaces should be assigned a unique IP address range.
- D. The Virtual-Access 1 interface must be enabled in an up/up state administratively

Answer: A

Explanation:

QUESTION NO: 17

Refer to the exhibit. The INSIDE zone has been configured and assigned to two separate router interfaces. All other zones and interfaces have been properly configured. Given the configuration example shown, what can be determined?

```
ip access-list extended INTRA_ZONE_ACL
  permit tcp 10.10.10.0 0.0.0.255 10.10.10.0 0.0.0.255 eq ssh
!
class-map type inspect INTRAZONE_CLASS
  match access-group name INTRA_ZONE_ACL
!
policy-map type inspect INTRAZONE_POLICY
  class type inspect INTRAZONE_CLASS
    inspect
  class class-default
    drop log
!
zone-pair security INTRAZONE source INSIDE destination INSIDE
service-policy type inspect INTRAZONE_POLICY
```

- A.** Hosts in the INSIDE zone, with addresses in the 10.10.10.0/24 network, can access any host in the 10.10.10.0/24 network using the SSH protocol.
- B.** If a host in the INSIDE zone attempts to communicate via SSH with another host on a different interface within the INSIDE zone, communications must pass through the router self zone using the INTRAZONE policy.
- C.** This is an illegal configuration. You cannot have the same source and destination zones.
- D.** This policy configuration is not needed, traffic within the same zone is allowed to pass by default.

Answer: B

Explanation:

The zone pair can also be configured to control the traffic permitted directly into the device; this includes control and management plane traffic. This is configured by creating a zone pair using the self zone as the source or destination zone. With the release of IOS 15.0.1M, it is also possible to control the traffic within the same zone; this is referred to as intrazone.

This is configured by creating a zone pair with the same two zone names as both source and destination.

QUESTION NO: 18

Which action does the command `private-vlan association 100,200` take?

- A. configures VLANs 100 and 200 and associates them as a community
- B. associates VLANs 100 and 200 with the primary VLAN
- C. creates two private VLANs with the designation of VLAN 100 and VLAN 200
- D. assigns VLANs 100 and 200 as an association of private VLANs

Answer: B

Explanation:

QUESTION NO: 19

Which of these allows you to add event actions globally based on the risk rating of each event, without having to configure each signature individually?

- A. event action summarization
- B. event action filter
- C. event action override
- D. signature event action processor

Answer: C

Explanation:

QUESTION NO: 20

When using Cisco Easy VPN, what are the three options for entering an XAUTH username and password for establishing a VPN connection from the Cisco Easy VPN remote router? (Choose three.)

- A. using an external AAA server
- B. entering the information via the router `crypto ipsec client ezvpn connect` CLI command in privileged EXEC mode
- C. using the router local user database
- D. entering the information from the PC via a browser
- E. storing the XAUTH credentials in the router configuration file

Answer: B,D,E

Explanation:

Begin by configuring the local network AAA authorization list with the `aaa authorization network` command.

This will tell the router to use only the locally configured user database on the router for its authorization resource.

C

If XAUTH is being used, it must be decided where to store the authentication credentials:

Store the XAUTH username and password in the configuration file on the router: This option is typically used if the router is shared between many PCs and the goal is to have the VPN tunnel up all the time.

E

Do not store the XAUTH username and password on the router: If this option is used, a PC user who is connected to the router is presented with a web page that allows the username and password to be manually entered.

D

EZVPN Remote connection profile using the `crypto ipsec client ezvpn` command

Use the `group` command to specify the group name and group password to authenticate to the EZVPN Server as a part of a group.

Use the `username` command to specify the stored username and password used to provide additional authentication using XAUTH.

B

QUESTION NO: 21

Which of these is true regarding tunnel configuration when deploying a Cisco ISR as a DMVPN hub router?

A. Only one tunnel can be created per tunnel source interface.

B. Only one tunnel can be created and should be associated with a loopback interface for dynamic redundancy

C. The GRE tunnel key is used to encrypt the traffic going through the tunnel through the hub.

D. You can run multiple parallel DMVPNs on the hub router, but each tunnel requires a unique tunnel key.

Answer: D

Explanation:

Task 4 creates the mGRE tunnel interface. Enter the `interface tunnel` command and then configure basic GRE parameters. The `tunnel mode gre multipoint` command designates the tunnel interface

as mGRE and the tunnel source command specifies the physical interface to which the GRE tunnel is bound. The tunnel key command is required and must match the tunnel key configured on the spokes. This command allows network administrators to run more than one DMVPN at a time on the same router. The GRE tunnel key therefore uniquely identifies the DMVPN.

QUESTION NO: 22

Given the Cisco IOS command `crypto key generate rsa label MY_KEYS modulus 2048`, which additional command keyword should be added if you would like to use these keys on another router or have the ability to back them up to another device?

- A. redundancy
- B. exportable
- C. on:USB smart-token
- D. usage-keys

Answer: B

Explanation:

QUESTION NO: 23

Which two types of deployments can be implemented for a zone-based policy firewall? (Choose two.)

- A. routed mode
- B. interzone mode
- C. fail open mode
- D. transparent mode
- E. inspection mode

Answer: A,D

Explanation:

QUESTION NO: 24 DRAG DROP

Drag the items on the left to the empty boxes on the right that best define Application Inspection and Control benefits. Not all items will be used.

dropping application layer protocol units that do not conform to the protocol standard

an application-aware method of filtering that works on OSI layers 3 and 4

in-memory reassembly of Layer 4 (TCP, UDP) sessions to obtain the sequential stream of bytes exchanged between hosts

filtering inside the protocol and its related content

Answer:

Drag the items on the left to the empty boxes on the right that best define Application Inspection and Control benefits. Not all items will be used.

dropping application layer protocol units that do not conform to the protocol standard

an application-aware method of filtering that works on OSI layers 3 and 4

in-memory reassembly of Layer 4 (TCP, UDP) sessions to obtain the sequential stream of bytes exchanged between hosts

filtering inside the protocol and its related content

dropping application layer protocol units that do not conform to the protocol standard

in-memory reassembly of Layer 4 (TCP, UDP) sessions to obtain the sequential stream of bytes exchanged between hosts

filtering inside the protocol and its related content

Explanation:

Drag the items on the left to the empty boxes on the right that best define Application Inspection and Control benefits. Not all items will be used.

an application-aware method of filtering that works on OSI layers 3 and 4

dropping application layer protocol units that do not conform to the protocol standard

in-memory reassembly of Layer 4 (TCP, UDP) sessions to obtain the sequential stream of bytes exchanged between hosts

filtering inside the protocol and its related content

QUESTION NO: 25

What is the result of configuring the command `dot1x system-auth-control` on a Cisco Catalyst switch?

- A. enables the switch to operate as the 802.1X supplicant
- B. globally enables 802.1X on the switch
- C. globally enables 802.1X and defines ports as 802.1X-capable
- D. places the configuration sub-mode into `dot1x-auth` mode, in which you can identify the authentication server parameters

Answer: B

Explanation:

QUESTION NO: 26

Which information is displayed when you enter the Cisco IOS command `show epm session`?

- A. Enforcement Policy Module sessions
- B. External Proxy Mappings, per authenticated sessions
- C. Encrypted Policy Management sessions
- D. Enhanced Protected Mode sessions

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/ios/security/command/reference/sec_s4.html#wp1063145

QUESTION NO: 27

Refer to the exhibit. Based on the partial configuration shown, which the GET VPN group member GDOI configuration?

```
!  
crypto gdoi group CLASSROOM  
identity number 12345  
!  
crypto map GETVPN_MAP 10 gdoi  
set group CLASSROOM  
!  
interface Serial0/0/0  
ip address 192.168.1.2 255.255.255.252  
crypto map GETVPN_MAP  
!
```

- A. key server IP address
- B. local priority
- C. mapping of the IPsec profile to the IPsec SA
- D. mapping of the IPsec transform set to the GDOI group

Answer: A

Explanation:

QUESTION NO: 28

Refer to the exhibit. Given the partial configuration shown, which two statements are correct? (Choose two.)

```
crypto ipsec transform-set MY_TRANSFORM esp-aes 128 esp-sha-hmac  
!  
crypto ipsec profile MYPROFILE  
set transform-set MY_TRANSFORM  
!  
interface Tunnel0  
ip unnumbered GigabitEthernet1/1  
tunnel source GigabitEthernet1/1  
tunnel destination 192.168.2.200  
tunnel mode ipsec ipv4  
tunnel protection ipsec profile MYPROFILE  
!  
ip route 10.1.2.0 255.255.255.0 Tunnel0
```

- A. The tunnel will use the routing protocol configured for GigabitEthernet 1/1 for all tunnel communication with the peer.

- B.** The IP route statement to reach the remote network behind the DMVPN peer is incorrect, it should be ip route 192.168.2.0 255.255.255.0 tunnel 0.
- C.** This is an example of a static point-to-point VTI tunnel.
- D.** The tunnel will use esp-sha-hmac encryption in ESP tunnel mode.
- E.** The tunnel will use 128-bit AES encryption in ESP tunnel mode.

Answer: C,E

Explanation:

QUESTION NO: 29

You are troubleshooting a Cisco Easy VPN installation that is experiencing session establishment problems. You have verified that matching IKE and IPsec policies exist on both peers. The remote client has also successfully entered authentication credentials. What is the next step to take in troubleshooting this problem?

- A.** verify that the router is not denying traffic from the tunnel
- B.** verify that the router is able to assign an IP address to the client
- C.** examine routing tables
- D.** issue a ping from the client to the router to verify reachability

Answer: B

Explanation:

QUESTION NO: 30

Which of these is a result of using the same routing protocol process for routing outside and inside the VPN tunnel?

- A.** This will provide for routing-protocol-based failover redundancy.
- B.** Spoke routers will be able to dynamically learn routes to peer networks.
- C.** This will allow VPN-encapsulated packets to be routed out the correct physical interface used to reach the remote peer
- D.** The tunnel will constantly flap.

Answer: D

Explanation:

Recursive Routing Hazard You must take precautions when configuring dynamic routing protocols to ensure that there is a device that participates in the same routing protocol both outside the VPN tunnel (the transport network) and inside the tunnel (directly with VPN peers).

This could be a possibility if an organization is in control of the transport network and wants to provide high availability through dynamic routing, both inside the transport network and inside the VPN to ensure continuous connectivity.

This kind of routing requires that VPN devices be prevented from learning the paths to their remote peer tunnel destination IP addresses over the VPN tunnel itself. The single hop path over the VPN will always be a better route than the path over the transport network. This situation will break the tunnel because it causes the VPN-encapsulated packet to be routed into its own tunnel interface instead of being routed out the correct physical interface that is used to reach the remote VPN peer. Cisco IOS Software will react to this behavior by flapping the tunnel interface.

Use either route filtering or a different routing protocol for the transport network and the VPN network to avoid this recursive routing issue.

QUESTION NO: 31 DRAG DROP

Drag the items on the left to the boxes on the right that represent features of Cisco Secure ACS when implementing 802.1X. Not all items will be used.

VLAN assignment	
ACL management	
provisioning for stateful packet filtering ACLs at the switch port level	
time-based access	
endpoint posture assessment	

Answer:

Drag the items on the left to the boxes on the right that represent features of Cisco Secure ACS when implementing 802.1X. Not all items will be used.

VLAN assignment	VLAN assignment
ACL management	time-based access
provisioning for stateful packet filtering ACLs at the switch port level	ACL management
time-based access	
endpoint posture assessment	

Explanation:

Drag the items on the left to the boxes on the right that represent features of Cisco Secure ACS when implementing 802.1X. Not all items will be used.

endpoint posture assessment

VLAN assignment

time-based access

provisioning for stateful packet filtering ACLs at the switch port level

ACL management

QUESTION NO: 32

Refer to the exhibit. What can be determined from the output of this show command?

```
switch#show interfaces Fa0/1 switchport
Name: fa0/1
Switchport: Enabled
...
Trunking VLANs Enabled: ALL
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: true
Appliance trust: none
```

- A. The switch port interface is enabled and operating as a community port.
- B. The interface is acting as an isolated switch port operating in VLAN 1.
- C. The interface is configured for Private VLAN Edge.
- D. The switch port interface is not a trusted port.

Answer: C**Explanation:**

QUESTION NO: 33

You are troubleshooting a problem related to IPsec connectivity issues. You see that there is no ISAKMP security association established between peers. You debug the connection process and see an error message of 1d00h: ISAKMP (0:1): atts are not acceptable. Next payload is 0. What does this message indicate?

- A. This indicates a policy mismatch.
- B. This indicates that the offered attributes did not contain a payload.
- C. IKE has failed initial attempts and will resend policy offerings to the peer router.
- D. The time stamp of the message shows that it is one day old. This could indicate a possible mismatch of system clocks and invalidate the connection attempt.

Answer: A

Explanation:

QUESTION NO: 34

Refer to the exhibit. Given the output shown, what can be determined?

```
%SW_DAI-4-ACL_DENY: 1 Invalid ARPs (Req) on Fa1/1, vlan 200.  
([0001.ba21.321c/192.168.1.10/0000.0000.0000/192.168.1.20/12:32:18  
UTC Mon Sep 20 2010])
```

- A. An attacker has sent a spoofed DHCP address.
- B. An attacker has sent a spoofed ARP response that violates a static mapping.
- C. The MAC address has matched a deny rule within the ACL.
- D. This is an invalid proxy ARP packet, as indicated by the 0000.0000.0000 MAC address on the destination

Answer: B

Explanation:

You can create an extended ACL with MAC address mapping.

If you have a spoofed arp then the message will be different than ACL-DENY - it will be DHCP Snooping Deny.

http://www.cisco.com/en/US/docs/switches/datacenter/sw/4_1/nx-os/security/configuration/guide/sec_arpinspect.html#wp1125009

3550(config-arp-nacl)#permit ip host 192.168.69.25 mac host 000c.2957.6b39 log

This will permit a host with an IP of 192.168.69.25 and a Mac of 00-0c-29-57-6b-39 to arp on the network.

If Host 2 attempts to send an ARP request with the IP address 10.0.0.1, DAI drops the request and logs the following system message:

00:18:08: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Ethernet1/4, vlan 1.([0001.0001.0001/10.0.0.1/0000.0000.0000/0.0.0.0/01:53:21 UTC Fri Jun 13 2008])

00:12:08: %SW_DAI-4-DHCP_SNOOPING_DENY: 2 Invalid ARPs (Req) on Ethernet2/3, vlan 1.([0002.0002.0002/10.0.0.3/0000.0000.0000/0.0.0.0/02:42:35 UTC Fri Jul 13 2008])

QUESTION NO: 35

Which command will enable a SCEP interface when you are configuring a Cisco router to be a certificate server?

- A. scep enable (under interface configuration mode)
- B. crypto pki scep enable
- C. grant auto
- D. ip http server

Answer: D

Explanation:

QUESTION NO: 36

When 802.1X is implemented, how do the client (supplicant) and authenticator communicate?

- A. RADIUS
- B. TACACS+
- C. MAB
- D. EAPOL

Answer: D

Explanation:

Note: EAPOL is used between the supplicant and the authenticator, while RADIUS is used between the authenticator and the authentication server.

QUESTION NO: 37

Refer to the exhibit. Assuming that all other supporting configurations are correct, what can be determined from the partial IP admission configuration shown?

```
aaa new-model
aaa authentication login default local
aaa authorization auth-proxy default local
!
aaa attribute list SUPERUSER-ATTRS
attribute type supplicant-group "SUPERUSER"
!
Username power1 privilege 15 secret thl3F4ftvA
username power1 aaa attribute list SUPERUSER-ATTRS
!
ip admission name USER-AUTH proxy http inactivity-time 60
ip http server
!
interface GigabitEthernet0/0
 ip admission USER-AUTH
 zone-member security OUTSIDE
```

- A. The router will forward authentication requests to a AAA server for authentication and authorization.
- B. The local user password is thl3F4ftvA.
- C. The router will intercept incoming HTTP sessions on interface G0/0 for authentication.
- D. The SUPERUSER's privilege level is being restricted.
- E. The attribute type supplicant-group "SUPERUSER" configuration can be used to match criteria in the "inspect" class-map type using the match access-group option.

Answer: C

Explanation:

QUESTION NO: 38

Which of these is an implementation guideline when deploying the IP Source Guard feature in an environment with multiple switches?

- A. Do not configure IP Source Guard on inter-switch links.
- B. Configure PACLS for DHCP-addressed end devices.
- C. IP Source Guard must be configured in the trunk sub-configuration mode to work on inter-switch links.
- D. Configure static IP Source Guard mapping for all access ports.

Answer: A

Explanation:

QUESTION NO: 39 DRAG DROP

Match the NAT types on the left with their definitions on the right.

Static Inside NAT

maps a local IP address in the inside NAT domain to a global IP address from a pool configured on the NAT gateway

Static Inside PAT

translates both the local IP address and source port of local hosts to distinguish between translations to a single global address in the outside domain

Dynamic Inside NAT

maps a specific local IP address in the inside NAT domain to another specific global IP address with a preconfigured rule

Dynamic Inside PAT

maps a local host IP address and service port in the inside NAT domain to a global IP address and global port

Answer:

Match the NAT types on the left with their definitions on the right.

Static Inside NAT

Dynamic Inside NAT

Static Inside PAT

Dynamic Inside PAT

Dynamic Inside NAT

Static Inside NAT

Dynamic Inside PAT

Static Inside PAT

Explanation:

Match the NAT types on the left with their definitions on the right.



Dynamic Inside NAT

Dynamic Inside PAT

Static Inside NAT

Static Inside PAT

QUESTION NO: 40

What does the command `errdisable recovery cause arp-inspection interval 300` provide for?

- A. It will disable a port when the ARP rate limit of 300 packets per second is received and wait a configured interval time before placing the port back in normal operation.
- B. It will inspect for ARP-disabled ports every 300 seconds.
- C. It will recover a disabled port and limit ARP traffic to 300 packets per second to avoid potential ARP attacks from reoccurring.
- D. It will recover a disabled port due to an ARP inspection condition in 5 minutes.

Answer: D

Explanation:

QUESTION NO: 41

You have configured Management Plane Protection on an interface on a Cisco router. What is the resulting action on implementing MPP?

- A. Inspection of protected management interfaces is automatically configured to ensure that management protocols comply with standards.
- B. The router gives preference to the configured management interface. If that interface becomes unavailable, management protocols will be allowed on alternate interfaces.
- C. Along with normal user data traffic, management traffic is also allowed only on the protected

interface.

D. Only management protocols are allowed on the protected interface.

Answer: D

Explanation:

QUESTION NO: 42 DRAG DROP

Drag the items on the left to the boxes on the right that represent mitigation techniques for VLAN hopping. Not all items will be used.

use BPDU port guard	
use static access ports	
disable DTP	
disable VTP	
avoid trunk native VLAN on access ports	

Answer:

Drag the items on the left to the boxes on the right that represent mitigation techniques for VLAN hopping. Not all items will be used.

use BPDU port guard	use static access ports
use static access ports	disable DTP
disable DTP	avoid trunk native VLAN on access ports
disable VTP	
avoid trunk native VLAN on access ports	

Explanation:

Drag the items on the left to the boxes on the right that represent mitigation techniques for VLAN hopping. Not all items will be used.

use BPDU port guard	use static access ports
	disable DTP
	avoid trunk native VLAN on access ports
disable VTP	

QUESTION NO: 43

Refer to the exhibit. What can be determined from the configuration shown?

```
snmp-server view SNMP-MGMT-VIEW interfaces included
!
snmp-server group SNMP-GROUP-MGMT v3 priv read SNMP-MGMT-VIEW access 90
snmp-server user SNMP-USER SNMP-GROUP-MGMT v3 auth sha
    Xbt4th33xAm priv aes 128 A12eS8v3Acc90
!

snmp-server host 10.0.100.2 traps traps version 3 auth SNMP-USER
snmp-server enable traps
```

- A. The community SNMP string is SNMP-MGMT-VIEW.
- B. All interfaces will be included in the SNMP GETs.
- C. This SNMP group will only allow read access to interface MIBs.
- D. The SNMP server group is using 128-bit SHA authentication.

Answer: C

Explanation:

First line -- interfaces included specifies that this view is only allowed to see the interface MIB's

QUESTION NO: 44

When enabling the Cisco IOS IPS feature, which step should you perform to prevent rogue signature updates from being installed on the router?

- A. configure authentication and authorization for maintaining signature updates
- B. install a known RSA public key that correlates to a private key used by Cisco
- C. manually import signature updates from Cisco to a secure server, and then transfer files from the secure server to the router
- D. use the SDEE protocol for all signature updates from a known secure management station

Answer: B

Explanation:

QUESTION NO: 45

A user has requested a connection to an external website. After initiating the connection, a message appears in the user's browser stating that access to the requested website has been denied by the company usage policy. What is the most likely reason for this message to appear?

- A. An antivirus software program has blocked the session request due to potential malicious content.
- B. The network has been configured with a URL filtering service.
- C. The network has been configured for 802.1X authentication and the user has failed to authenticate
- D. The user's configured policy access level does not contain proper permissions

Answer: B

Explanation:

QUESTION NO: 46

Refer to the exhibit. Given the partial configuration shown, what can be determined.

```
ip local pool PAT-POOL 209.165.2.100
ip access-list extended NAT-ACL
 permit ip 10.10.30.0 0.0.0.255 10.100.100.0 0.0.0.255
ip nat inside source list NAT-ACL pool PAT-POOL overload
```

- A. This is an example of a dynamic policy PAT rule.
- B. This is an example of a static policy NAT rule.
- C. Addresses in the 10.10.30.0 network will be exempt from translation when destined for the 10.100.100.0 network.
- D. The extended access list provides for one-to-one translation mapping of the 10.10.30.0 network to the 10.100.100.0 network

Answer: A

Explanation:

QUESTION NO: 47

When is it most appropriate to choose IPS functionality based on Cisco IOS software?

- A. when traffic rates are low and a complete signature is not required
- B. when accelerated, integrated performance is required using hardware ASIC-based IPS inspections
- C. when integrated policy virtualization is required
- D. when promiscuous inspection meets security requirements

Answer: A

Explanation:

QUESTION NO: 48

When performing NAT, which of these is a limitation you need to account for?

- A. exhaustion of port number translations
- B. embedded IP addresses
- C. security payload identifiers
- D. inability to provide mutual connectivity to networks with overlapping address spaces

Answer: B

Explanation:

QUESTION NO: 49 DRAG DROP

Drag the items on the left to the boxes on the right that define security protection features of the control plane on Cisco Layer 2/3 multilayer Catalyst switches. Not all items will be used.

routing protocol filtering	
BPDU Guard	
port security	
VTP authentication	
routing protocol authentication	
VLAN ACLs	

Answer:

Drag the items on the left to the boxes on the right that define security protection features of the control plane on Cisco Layer 2/3 multilayer Catalyst switches. Not all items will be used.

routing protocol filtering	routing protocol filtering
BPDU Guard	BPDU Guard
port security	VTP authentication
VTP authentication	routing protocol authentication
routing protocol authentication	
VLAN ACLs	

Explanation:

Drag the items on the left to the boxes on the right that define security protection features of the control plane on Cisco Layer 2/3 multilayer Catalyst switches. Not all items will be used.

	routing protocol filtering
	BPDU Guard
port security	VTP authentication
	routing protocol authentication
VLAN ACLs	

QUESTION NO: 50

You have enabled Cisco IOS IPS on a router in your network. However, you are not seeing expected events on your monitoring system (such as Cisco IME). On the router, you see events being captured. What is the next step in troubleshooting the problem?

- A. verify that syslog is configured to send events to the correct server
- B. verify SDEE communications
- C. verify event action rules
- D. verify that the IPS license is valid

Answer: B

Explanation:

QUESTION NO: 51

Which two of these are features of control plane security on a Cisco ISR? (Choose two.)

- A. CoPP
- B. RBAC
- C. AAA
- D. CPPr
- E. uRPF
- F. FPM

Answer: A,D

Explanation:

QUESTION NO: 52

Which two of these are potential results of an attacker performing a DHCP server spoofing attack?
(Choose two.)

- A. DHCP snooping
- B. DoS
- C. confidentiality breach
- D. spoofed MAC addresses
- E. switch ports being converted to an untrusted state

Answer: B,C

Explanation:

QUESTION NO: 53

When Cisco IOS IPS signatures are being tuned, how is the Target Value Rating assigned?

- A. It is calculated from the Event Risk Rating.
- B. It is calculated from a combination of the Attack Severity Rating and Signature Fidelity Rating
- C. It is manually set by the administrator.
- D. It is set based upon SEAP functions.

Answer: C

Explanation:

QUESTION NO: 54

Which of these should you do before configuring IP Source Guard on a Cisco Catalyst switch?

- A. Enable NTP for event correlation
- B. Enable IP routing authentication
- C. Configure an access list with exempt DHCP-initiated IP address ranges
- D. Turn DHCP snooping on at least 24 hours in advance

Answer: D

Explanation:

QUESTION NO: 55

What action will the parameter-map type ooo global command enable?

- A. globally initiates tuning of the router's TCP normalizer parameters for out-of-order packets
- B. globally classifies type ooo packets within the parameter map and subsequent policy map
- C. enables a parameter map named ooo
- D. configures a global parameter map for traffic destined to the router itself

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/ios-xml/ios/sec_data_zbf/configuration/12-4t/sec-zone-pol-fw.html

QUESTION NO: 56 DRAG DROP

Drag the items on the left to the boxes on the right that define security protection features of the data plane on Cisco Catalyst switches. Not all items will be used.

AAA
port ACLs
port security
VLAN ACLs
RBAC
private VLANs

Answer:

Drag the items on the left to the boxes on the right that define security protection features of the data plane on Cisco Catalyst switches. Not all items will be used.

AAA	port ACLs
port ACLs	port security
port security	VLAN ACLs
VLAN ACLs	private VLANs
RBAC	
private VLANs	

Explanation:

Drag the items on the left to the boxes on the right that define security protection features of the data plane on Cisco Catalyst switches. Not all items will be used.

AAA	port ACLs
	port security
	VLAN ACLs
	private VLANs
RBAC	

untitled

QUESTION NO: 57 CORRECT TEXT

Scenario

To access the router console ports, refer to the exhibit and click router R1 or R2 for access. The ISP router will not need to be accessed. You will be placed directly into Cisco IOS enable mode. Using CLI `show` commands, answer the questions presented regarding GET VPN configurations. For this exercise, you will not be able to use `show running-config` or `show start-config` CLI commands. You may use other specific `show run` and global `show` commands to determine the correct answers. Not all Cisco IOS CLI features are enabled for this simulation.

Questions

0% Complete

1
2
3
4
5

Topology

ISP router

R1

R2

Question 1

What is the registration status of the group member router and what is the ip addresses of the key server? (Choose two.)

- ☐ group registration has not yet been attempted
- ☒ the member router is registered with the key server
- ☐ 192.168.2.2
- ☒ 192.168.1.2

Answer: R1# show crypto gdoi -or- R2# show crypto gdoi

Explanation:

This command will show you the KS ip address and your registration - with time to re-key

R1#show crypto gdoi

GROUP INFORMATION

Group Name: GETVPNGROUP

Group Identity: 67890

Rekeys received: 0

IPSec SA Direction: Both

Active Group Server: 192.168.1.2

Group Server list: 192.168.1.2

GM Reregisters in: 3434 secs

Rekey Received: never

Rekeys received

Cumulative: 0

After registration: 0

ACL Downloaded From KS 192.168.1.2:

access-list permit ip 0.0.0.0 255.255.255.0 0.0.0.0 255.255.255.0

TEK POLICY for the current KS-Policy ACEs Downloaded:

FastEthernet0/0:

IPsec SA:

spi: 0x673C7398(1732015000)

transform: esp-aes esp-sha-hmac

sa timing:remaining key lifetime (sec): (3571)

Anti-Replay: Disabled

QUESTION NO: 58 CORRECT TEXT

The screenshot shows a Cisco exam simulation interface. At the top, a 'Scenario' box contains instructions: 'To access the router console ports, refer to the exhibit and click: router R1 or R2 for access. The ISP router will not need to be accessed. You will be placed directly into Cisco IOS enable mode. Using CLI **show** commands, answer the questions presented regarding GET VPN configurations. For this exercise, you will not be able to use **show running-config** or **show start-config** CLI commands. You may use other specific **show run** and global **show** commands to determine the correct answers. Not all Cisco IOS CLI features are enabled for this simulation.'

On the left, a 'Questions' panel shows a list of questions (1-5) and a progress indicator at '0% Complete'. The main area displays a network topology with three routers: 'ISP router' at the top, 'R1' on the left, and 'R2' on the right. They are connected in a triangle. A 'Question 2' window is overlaid on the topology, asking: 'On the key server router, what is the name of the transform set applied to the IPsec profile and which protection services is the transform set providing? (Choose two.)' The options are:

- ☐ the name is ESP-3DES-SHA
- ☒ the name is GETSET
- ☒ the transform set is offering esp-aes esp-sha-hmac
- ☐ the transform set is offering esp-3des esp-sha-hmac

At the bottom, there are tabs for 'Scenario' and 'Topology'.

Answer: R2# show crypto ipsec transform-set

Explanation:

NB - only show runn commands accepted are show runn interfaces

R2#show crypto ipsec transform-set

Transform set GETSET: { esp-sha-hmac }

will negotiate = { Tunnel, },

{ esp-256-aes }

will negotiate = { Tunnel, },

!

QUESTION NO: 59 CORRECT TEXT

The screenshot displays a Cisco exam simulation interface. At the top, a 'Scenario' box contains instructions: 'To access the router console ports, refer to the exhibit and click router R1 or R2 for access. The ISP router will not need to be accessed. You will be placed directly into Cisco IOS enable mode. Using CLI **show** commands, answer the questions presented regarding GET VPN configurations. For this exercise, you will not be able to use **show running-config** or **show start-config** CLI commands. You may use other specific **show run** and global **show** commands to determine the correct answers. Not all Cisco IOS CLI features are enabled for this simulation.'

On the left, a 'Questions' sidebar shows a progress bar at '0% Complete' and a list of questions numbered 1 through 5, with question 3 currently selected.

The main area shows a network topology with three routers: 'ISP router' at the top, 'R1' on the left, and 'R2' on the right. A 'Question 3' dialog box is overlaid on the topology, asking: 'Which router is acting as the key server and which router is acting as a group member? (Choose two.)' The options are:

- ☐ Router 1 is the key server
- ☒ Router 2 is the key server
- ☒ Router 1 is the group member
- ☐ Router 2 is the group member
- ☐ The ISP router is the key server
- ☐ The ISP router is the group member
- ☐ Router 1 and Router 2 are both key servers

At the bottom, there are tabs for 'Scenario' and 'Topology'.

Answer: R2# show crypto gdoi ks -or- R2# show crypto gdoi ks members -or- R1# show ip interface brief

Explanation:

NB: it is assumed that only R1 is a member router and ISP is not a member

R1#show crypto gdoi ks

Total group members registered to this box: 0

Confirmed this is not the key server

R2#show crypto gdoi ks

Total group members registered to this box: 2

Key Server Information For Group GETVPNGROUP:

Group Name : GETVPNGROUP
Group Identity : 67890
Group Members : 1
IPSec SA Direction : Both
ACL Configured:
access-list 101

R2#show crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group GETVPNGROUP : 0

Group Member ID : 192.168.2.1
Group ID : 67890
Group Name : GETVPNGROUP
Key Server ID : 0.0.0.0

Confirm the IP address is associated with R1 and not ISP

R1#show ip interface brief

Interface	IP-Address	OK? Method Status	Protocol
FastEthernet0/0	192.168.2.1	YES manual up	up

All commands can be referenced here

http://www.cisco.com/en/US/docs/ios/security/command/reference/sec_s3.html#wp1159252

QUESTION NO: 60 CORRECT TEXT

Answer: R2# show crypto gdoi group GETVPNGROUP

Explanation:

R2 is better as this is the KS

R2#show crypto gdoi group GETVPNGROUP

Group Name: GETVPNGROUP (Multicast)

Group Identity: 67890

Group Members: 2

IPSec SA Direction: Both
Active Group Server: Local
Group Rekey Lifetime: 86400 secs
Rekey Retransmit Period: 10 secs
Rekey Retransmit Attempts: 2

IPSec SA Number: 10
IPSec SA Rekey Lifetime: 3600 secs
Profile Name: GETPROFILE
Replay method: Count Based
Replay Window Size: 64
SA Rekey
Remaining Lifetime: 1998 secs
ACL Configured: access-list 101
Group Server list: Local

NB: some other tests have 2 answers highlighted- the question does not ask for (Choose Two) and must assume on one selection is correct.

QUESTION NO: 61 CORRECT TEXT

Questions

0% Complete

1
2
3
4
5

Scenario

To access the router console ports, refer to the exhibit and click router R1 or R2 for access. The ISP router will not need to be accessed. You will be placed directly into Cisco IOS enable mode. Using CLI **show** commands, answer the questions presented regarding GET VPN configurations. For this exercise, you will not be able to use **show running-config** or **show start-config** CLI commands. You may use other specific **show** run and global **show** commands to determine the correct answers. Not all Cisco IOS CLI features are enabled for this simulation.

Actual

ISP router

R1

R2

On the group member router, where is the crypto map applied and what is the ISAKMP shared key? (Choose two.)

- ☒ The crypto map is applied to the FastEthernet0/1 interface
- ☐ The crypto map name is applied globally on the router and is active on all enabled interfaces
- ☒ The shared key is GETVPNKEY
- ☐ The shared key is 57890

Scenario **Topology**

Answer: R1# show crypto map -or- R1# show crypto isakmp key

Explanation:

R1 is the only group member that you can access so it is assumed this is the only group member

R1#show crypto map

Crypto Map "CMAP" 10 gdoi

Group Name: GETVPNGROUP

identity number 67890

server address ipv4 192.168.1.2

Interfaces using crypto map CMAP:

FastEthernet0/1

R1#show crypto isakmp key

Keyring	Hostname/Address	Preshared Key
default	192.168.1.2	GETVPNKEY

QUESTION NO: 62

Which protocol is EAP encapsulated in for communications between the authenticator and the authentication server?

- A. EAP-MD5
- B. IPsec
- C. EAPOL
- D. RADIUS

Answer: D

Explanation:

Note: EAPOL is used between the supplicant and the authenticator, while RADIUS is used between the authenticator and the authentication server.

QUESTION NO: 63

You are loading a basic IPS signature package onto a Cisco router. After a period of time, you see this message:

%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 275013 ms. What do you expect happened during downloading and compilation of the files?

- A.** The files were successfully copied with an elapse time of 275013 ms. The router will continue with extraction and compilation of the signature database.
- B.** The signature engines were compiles, but there is no indication that the actual signatures were compiled.
- C.** The compilation failed for some of the signature engines. There are 16 engines, but only 6 were completed according to the %IPS-6 message
- D.** The files were compiled without error.

Answer: D

Explanation:

http://www.cisco.com/en/US/prod/collateral/iosswrel/ps6537/ps6586/ps6634/prod_white_paper0900aecd805c4ea8.html

QUESTION NO: 64

Refer to the exhibit. Given the configuration shown, which of these statements is correct?


```
parameter-map type trend-global SERVICE-PARAMS
server trps1.trendmicro.com retrans 5 timeout 200
!
parameter-map type urlfpolicy trend URL-PARAMS
block-page message "ACCESS BLOCKED BY POLICY"
allow-mode on
!
class-map type urlfilter trend BAD-CONTENT
match url category Gambling
!
policy-map type inspect urlfilter URL-POLICY
parameter type urlfpolicy trend URL-PARAMS
class type urlfilter BAD-CONTENT
reset
!
class-map type inspect match-all WEB-CLASS
match protocol http
!
policy-map type inspect IN-TO-OUT-POLICY
class type inspect WEB-TRAFFIC
inspect
service-policy urlfilter URL-POLICY
!
zone-pair security IN-TO-OUT source INSIDE destination
OUTSIDE
service-policy type inspect IN-TO-OUT-POLICY
```

- A. An external service is providing URL filtering via a subscription service.
- B. All HTTP traffic to websites with the name "Gambling" included in the URL will be reset.
- C. A service policy on the zone pair needs to be configured in the opposite direction or all return HTTP traffic will be blocked by policy
- D. The URL filter policy has been configured in a fail-closed scenario.

Answer: A

Explanation:

QUESTION NO: 65 DRAG DROP

Drag the items on the left to the boxes on the right that represent the initial state of a DMVPN deployment. Not all items will be used.

NHRP mappings are created.

Spoke router route tables are updated with peer IP addresses.

All spoke traffic is forwarded to the hub.

Spoke-to-hub GRE and IPsec tunnels are created.

Spoke-to-spoke GRE and IPsec unidirectional tunnels are created.

Answer:

Drag the items on the left to the boxes on the right that represent the initial state of a DMVPN deployment. Not all items will be used.

NHRP mappings are created.

Spoke router route tables are updated with peer IP addresses.

All spoke traffic is forwarded to the hub.

Spoke-to-hub GRE and IPsec tunnels are created.

Spoke-to-spoke GRE and IPsec unidirectional tunnels are created.

NHRP mappings are created.

Spoke-to-hub GRE and IPsec tunnels are created.

All spoke traffic is forwarded to the hub.

Explanation:

Drag the items on the left to the boxes on the right that represent the initial state of a DMVPN deployment. Not all items will be used.

Spoke router route tables are updated with peer IP addresses.

NHRP mappings are created.

Spoke-to-hub GRE and IPsec tunnels are created.

All spoke traffic is forwarded to the hub.

Spoke-to-spoke GRE and IPsec unidirectional tunnels are created.

Page 453 - CCNP Security Guide - Initial State

In its initial state, the network is purely hub-and-spoke and can stay that way if desired.

The initial network properties are:

QUESTION NO: 66

Refer to the exhibit. Which two of these are most likely to have caused the issue with NHRP, given this output of the show command? (Choose two.)

```
Spoke#show ip nhrp nhs detail
```

```
Legend: E=Expecting replies, R=Responding
```

```
Tunnel0: 192.168.1.1 E req-sent 0 req-failed 120 repl-recvd 0
```

```
Pending Registration Requests:
```

```
Registration Request: Reqid 1201, Ret 64 NHS 192.168.1.1
```

- A. There was a network ID mismatch.
- B. The spoke router has not yet sent a request via Tunnel0.
- C. The spoke router received a malformed NHRP packet.
- D. There was an authentication key mismatch.
- E. The registration request was expecting a return request ID of 1201, but received an ID of 120.

Answer: A,D

Explanation:

QUESTION NO: 67 DRAG DROP

Match the Cisco IOS IPS features on the left with their descriptions on the right. Not all features will be used.

risk rating

target value rating

signature fidelity rating

event action override

alert severity rating

event action filter

removes actions from an event

adds event actions globally

represents the perceived value of a target host

Answer:

Match the Cisco IOS IPS features on the left with their descriptions on the right. Not all features will be used.

risk rating	event action filter
target value rating	event action override
signature fidelity rating	target value rating
event action override	
alert severity rating	
event action filter	

Explanation: Match the Cisco IOS IPS features on the left with their descriptions on the right. Not all features will be used.

risk rating	event action filter
	event action override
signature fidelity rating	target value rating
alert severity rating	

QUESTION NO: 68

You have configured a guest VLAN using 802.1X on a Cisco Catalyst switch. A client incapable of using 802.1X has accessed the port and has been assigned to the guest VLAN. What happens when a client capable of using 802.1X joins the network on the same port?

- A.** The client capable of using 802.1X is allowed access and proper security policies are applied to the client.
- B.** EAPOL packets will not be allowed on the guest VLAN and the access attempt will fail.
- C.** The port is put into the unauthorized state in the user-configured access VLAN, and authentication is restarted.
- D.** This is considered a security breach by the authentication server and all users on the access port will be placed into the restricted VLAN.

Answer: C**Explanation:**

Usage Guidelines for Using Authentication Failed VLAN Assignment

When an authentication failed port is moved to an unauthorized state the authentication process is restarted. If you should fail the authentication process again the authenticator waits in the held

state. After you have correctly reauthenticated all 802.1x ports are reinitialized and treated as normal 802.1x ports.

<http://www.cisco.com/en/US/docs/switches/lan/catalyst4500/12.2/31sga/configuration/guide/dot1x.html#wp1198927>

QUESTION NO: 69

Refer to the exhibit. What can be determined from the information shown?

```
Router# show access-list
Standard IP access list 10
 10 permit 10.0.100.0 0.0.0.255
Router# show ip eigrp route
^
% Invalid input detected at '^' marker.
Router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# enable secret mycisco
^
% Invalid input detected at '^' marker.
Router(config)# ip access-list extended DMZ_ACL
Router(config-ext-nacl)#
```

- A. The user has been restricted to privilege level 1.
- B. The standard access list should be reconfigured as an extended access list to allow desired user permissions
- C. RBAC has been configured with restricted views.
- D. IP access list DMZ_ACL has not yet been configured with proper permissions.

Answer: C

Explanation:

QUESTION NO: 70

Refer to the exhibit. Assuming that all other supporting configurations are correct, what can be determined from the partial IP admission configuration shown?

```
aaa new-model
aaa authentication login default local
aaa authorization auth-proxy default local
!
aaa attribute list ADMIN_LIST
attribute type supplicant-group "ADMINISTRATOR"
!
username maint3nanc3 privilege 7 secret Hra44GiA0b
username maint3nanc3 aaa attribute list ADMIN_LIST
!
ip admission name ADMIN_AUTH proxy http inactivity-time 20
!
interface GigabitEthernet0/1
 ip admission ADMIN_AUTH
 zone-member security INSIDE
```

- A. The router will forward authentication requests to a AAA server for authentication and authorization.
- B. The user maint3nanc3 will have complete CLI command access once authenticated.
- C. After a period of 20 minutes, the user will again be required to provide authentication credentials.
- D. The authentication proxy will fail, because the router's HTTP server has not been enabled.
- E. All traffic entering interface GO/1 will be intercepted for authentication, but only Telnet traffic will be authorized.

Answer: C

Explanation:

QUESTION NO: 71

What will the authentication event fail retry 0 action authorize vlan 300 command accomplish?

- A. assigns clients that fail 802.1X authentication into the restricted VLAN 300
- B. assigns clients to VLAN 300 and attempts reauthorization
- C. assigns a client to the guest VLAN 300 if it does not receive a response from the client to its EAPOL request/identity frame

D. locks out a user who fails an 802.1X authentication and does not allow the user to try to gain network access again for 300 seconds

Answer: A

Explanation:

QUESTION NO: 72 DRAG DROP

Match the Application Inspection Control approaches on the left with their descriptions on the right.

protocol minimization	can prevent covert tunneling
payload minimization	increases protection by only allowing expected content types and values
application layer inspections	increases protection by hiding unnecessary features
protocol verification	can be configured in a zone-based firewall policy to prevent known attacks

Answer:

Match the Application Inspection Control approaches on the left with their descriptions on the right.

protocol minimization	protocol verification
payload minimization	payload minimization
application layer inspections	protocol minimization
protocol verification	application layer inspections

Explanation:

Match the Application Inspection Control approaches on the left with their descriptions on the right.

	protocol verification
	payload minimization
	protocol minimization
	application layer inspections

<http://www.slideshare.net/CiscoSystems/ccsp-effective-deployment-of-cisco-asa-access-control>

QUESTION NO: 73

When you are configuring a DMVPN network, which tunnel mode should you use for the hub router configuration?

- A. GRE multipoint
- B. Nonbroadcast multiaccess
- C. Classic point-to-point GRE
- D. IPsec multipoint

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/solutions/Enterprise/WAN_and_MAN/DMVPN_2_Phase2.html

The hub-and-spoke deployment model is the most common deployment model. This model is the most scalable, and predominately mimics traditional Layer 2 leased line, Frame Relay, or ATM hub-and-spoke networks. The headend is configured with a multipoint GRE (mGRE) interface, and the branch with a point-to-point (p2p) GRE interface.

QUESTION NO: 74

Which Cisco IOS feature provides secure, on-demand meshed connectivity?

- A. DMVPN
- B. Easy VPN
- C. IPsec VPN
- D. mGRE

Answer: A

Explanation:

QUESTION NO: 75

You have configured a Cisco router to act a PKI certificate server. However, you are experiencing problems starting the server. You have verified that all CA parameters have been correctly configured. What is the next step you should take in troubleshooting this problem?

- A. Disable and restart the router's HTTP server function
- B. Enable the SCEP interface
- C. Verify the RSA key pair and generate new keys
- D. Verify that correct time is being used and source are reachable

Answer: D

Explanation:

There are others who prefer the answer from the previous dump.

However, the question clearly states "You have verified that all CA parameters have been correctly configured"

So if the configuration is correctly configured, why would you enable SCEP interface again? The best answer is verify correct time is being used and source are reachable

Having synchronized time is vital for PKI, but PKI does not require that the time be extremely accurate.

Time synchronization issues can cause certificate validation failures if the current time on the VPN device is outside the validity range of the CA certificate.

QUESTION NO: 76

Which three of these are features of data plane security on a Cisco ISR? (Choose three)

- A. uRPF
- B. NetFlow export
- C. FPM
- D. CPPr
- E. RBAC
- F. routing protocol filtering

Answer: A,B,C

Explanation:

<http://ptgmedia.pearsoncmg.com/images/9781587142802/samplepages/1587142805.pdf>

QUESTION NO: 77

What will the authentication event fail retry 0 action authorize vlan 300 command accomplish?

- A. assigns clients that fail 802.1X authentication into the restricted VLAN 300
- B. assigns clients to VLAN 300 and attempts reauthorization
- C. assigns a client to the guest VLAN 300 if it does not receive a response from the client to its EAPOL request/identity frame
- D. locks out a user who fails an 802.1X authentication and does not allow the user to try to gain network access again for 300 seconds

Answer: A

Explanation:

QUESTION NO: 78

When you are configuring DHCP snooping, how should you classify access ports?

- A. untrusted
- B. trusted
- C. promiscuous
- D. private

Answer: A

Explanation:

QUESTION NO: 79

When configuring URL filtering with the Trend Micro filtering service. Which of these steps must you take to prepare for configuration?

- A. define blacklists and whitelists
- B. categorize traffic types
- C. install the appropriate root CA certificate on the router
- D. synchronize clocks via NTP to ensure accuracy of URL filter updates from the service

Answer: B

Explanation:

http://www.cisco.com/en/US/prod/collateral/iosswrel/ps6537/ps6586/ps6643/white_paper_c89-492776.pdf

QUESTION NO: 80

Which of these is correct regarding the functionality of DVTI tunnels?

- A.** DVTI tunnels are created dynamically from a preconfigured template as tunnels are established to the hub.
- B.** The hub router needs a static DVT1 tunnel to each spoke router in order to establish remote communications from spoke to spoke.
- C.** Spoke routers require a virtual template to clone the configuration on which the DVTI tunnel is established.
- D.** DVTI tunnels appear on the hub as tunnel interfaces.

Answer: A

Explanation:

http://www.cisco.com/en/US/docs/ios/12_3t/12_3t14/feature/guide/gtIPScrm.pdf

QUESTION NO: 81

When implementing GET VPN, which of these is a characteristic of GDOI IKE?

- A.** GDOI IKE sessions are established between all peers in the network.
- B.** Security associations do not need to linger between members once a group member has authenticated to the key server and obtained the group policy.
- C.** Each pair of peers has a private set of IPsec security associations that is only shared between the two peers.
- D.** GDOI IKE uses UDP port 500.

Answer: B

Explanation:

http://www.cisco.com/en/US/prod/collateral/iosswrel/ps6537/ps6586/ps6635/ps7180/deployment_guide_c07_554713.pdf

QUESTION NO: 82 DRAG DROP

Match the Items on the left with the VPN categories on the right.

Mannual provisioning of paths	User Traffic Encapsulation
Tunneling VPN	
PKI provisinging	
Non-Tunneling VPN	Configuration Scalability
Mannual Provisioning of Peer Identity	
Automated Peer Discovery	
	Authentication Scalability

Answer:

Match the Items on the left with the VPN categories on the right.

Mannual provisioning of paths	User Traffic Encapsulation
Tunneling VPN	Tunneling VPN
PKI provisinging	Non-Tunneling VPN
Non-Tunneling VPN	Configuration Scalability
Mannual Provisioning of Peer Identity	Mannual provisioning of paths
Automated Peer Discovery	Automated Peer Discovery
	Authentication Scalability
	Mannual Provisioning of Peer Identity
	PKI provisinging

Explanation:

Match the Items on the left with the VPN categories on the right.

User Traffic Encapsulation
Tunneling VPN
Non-Tunneling VPN
Configuration Scalability
Manual provisioning of paths
Automated Peer Discovery
Authentication Scalability
Manual Provisioning of Peer Identity
PKI provisioning

QUESTION NO: 83 DRAG DROP

Drag the items on the left to the boxes on the right in the sequence used for Cisco Easy VPN connection initiation.

The VPN client establishes an ISAKMP SA	
The mode configuration process is initiated	
IPsec quick mode completes the connection process	
The VPN client initiates IKE Phase 1	
The PRI process is initiated	
The Cisco easy VPN server accepts the SA protocol	
The Cisco easy VPN server initiates a username and password challenge	

Answer:

Drag the items on the left to the boxes on the right in the sequence used for Cisco Easy VPN connection initiation.

The VPN client establishes an ISAKMP SA	The VPN client initiates IKE Phase 1
The mode configuration process is initiated	The VPN client establishes an ISAKMP SA
IPsec quick mode completes the connection process	The Cisco easy VPN server accepts the SA protocol
The VPN client initiates IKE Phase 1	The Cisco easy VPN server initiates a username and password challenge
The PRI process is initiated	The mode configuration process is initiated
The Cisco easy VPN server accepts the SA protocol	The PRI process is initiated
The Cisco easy VPN server initiates a username and password challenge	IPsec quick mode completes the connection process

Explanation:

- Step 1 – The VPN Client initiates IKE Phase 1.
- Step 2 – The VPN Client establishes an ISAKMP SA.
- Step 3 – The Easy VPN Server accepts the SA proposal.
- Step 4 – The Easy VPN Server initiates a username and password challenge.
- Step 5 – The mode configuration process is initiated.
- Step 6 – The RRI process is initiated.
- Step 7 – IPSec quick mode completes the connection process

QUESTION NO: 84

Which of these are the two types of keys used when implementing GET VPN? (Choose two)

- A. public key
- B. group encryption
- C. traffic encryption key
- D. pre-shared key
- E. key encryption
- F. private key

Answer: C,E

Explanation:

QUESTION NO: 85 CORRECT TEXT

Scenario:

You have been given the task of performing initial zone-based policy firewall configurations. You will need to create zones, assign the zones to specific interfaces, and create zone pairs to allow for traffic flow between interfaces. You will also need to define a zone-based policy firewall and assign the policy to the zone pair. To access the router console ports, refer to the exhibit, click the router for access, and perform the following tasks.



Note that when performing the configuration, you should use the exact names highlighted in bold below:

Globally create zones and label them with the following names:

- OUTSIDE
- IHSIDE
- Assign interfaces to zones as indicated in the exhibit
- Create a zone pair for traffic flowing from the inside to outside zones named IH-TO-OUT - • Define a zone-based firewall policy named IH-TO-OUT-POLICY
- Use the "match protocol" classification option to statefully inspect HTTP traffic and drop all other traffic
- Use a class-map named HTTP_POLICY

Apply zone-based firewall policy IN-TO-OUT-POLICY to the zone pair

Answer: First we divide the networks into 2 zones: Inside and Outside.

```
Router(config)#zone security INSIDE
Router(config)#zone security OUTSIDE
Router(config)#interface fa0/0/1
Router(config-if)#no shutdown
Router(config-if)#zone-member security INSIDE
Router(config)#interface fa0/0/0
Router(config-if)#no shutdown
Router(config-if)#zone-member security OUTSIDE
Router(config)#class-map type inspect match-any HTTP_POLICY
```

```
Router(config-cmap)#match protocol http
Router(config)#policy-map type inspect IN-TO-OUT-POLICY
Router(config-pmap)#class type inspect HTTP_POLICY
Router(config-pmap-c)#inspect
Router(config)#zone-pair security IN-TO-OUT-POLICY source INSIDE destination OUTSIDE
Router(config-sec-zone-pair)#service-policy type inspect IN-TO-OUT-POLICY
```

QUESTION NO: 86

Refer to the exhibit.

```
%SW_DAI-4-ACL_DENY: 1 Invalid ARPs (Req) on Fa1/1, vlan 200.
([0001.ba21.321c/192.168.1.10/0000.0000.0000/192.168.1.20/12:32:18
UTC Mon Sep 20 2010])
```

What can be determined from the partial configuration shown?

- A. The zone-based policy firewall is providing for bridging of non-IP protocols.
- B. Since the interfaces are in the same bridge group, access policies are not required.
- C. Traffic flow will be allowed to pass between the interfaces without being inspected.
- D. The zone-based policy firewall is operating in transparent mode.

Answer: D

Explanation:

QUESTION NO: 87

When is it feasible for a port to be both a guest VLAN and a restricted VLAN?

- A. this configuration scenario is never be implemented
- B. when you have configured the port for promiscuous mode
- C. when private VLANs have been configured to place each end device into different subnets
- D. when you want to allow both types of users the same services

Answer: D

Explanation:

QUESTION NO: 88

Refer to the exhibit.

```
SYSTEM IMAGE
-----
Image type                : Release
  Signer Information
    Common Name           : CiscoSystems
    Organization Unit      : C2900
    Organization Name      : CiscoSystems
  Certificate Serial Number : 4AC3507C
  Hash Algorithm           : SHA512
  Signature Algorithm      : 2048-bit RSA
  Key Version              : A

  Verifier Information
    Verifier Name          : ROMMON 1
    Verifier Version       : System Bootstrap, Version
                           15.0(1r)M1, RELEASE SOFTWARE (fc1)
```

What can be determined from the information provided in the system image output?

- A. The router supports LDAP.
- B. A Key Version of "A" indicates that this is an advanced IP security image of the Cisco IOS system.
- C. The router is in ROM monitor mode.
- D. This is a digitally-signed Cisco IOS image.

Answer: D

Explanation:

QUESTION NO: 89

Which three of these are sources used when the router is configured for URL filtering? (Choose three.)

- A. Websense URL filter
- B. AAA server downloadable ACLs
- C. ASA URL filter feature set
- D. Trend Micro cloud-based URL filter service
- E. locally configured filter rules on the router
- F. Cisco SenderBase URL filtering service

Answer: A,D,E

Explanation:

QUESTION NO: 90

In an 802.1X environment, which feature allows for non-802.1X-supported devices such as printers and fax machines to authenticate?

- A. multiauth
- B. WebAuth
- C. MAB
- D. 802.1X guest VLAN

Answer: C

Explanation:

QUESTION NO: 91

The advantages of virtual tunnel interfaces (VTIs) over GRE VPN solutions are which three of the following? (Choose three.)

- A. VTI can support QoS.
- B. VTI provides a routable interface.
- C. VTI supports nonencrypted tunnels.
- D. VTI is more scalable than a GRE-based VPN solution.
- E. IPsec VTIs need fewer established SAs to cover different types of traffic, both unicast and multicast, thus enabling improved scaling.
- F. IPsec VTIs require a loopback interface for configuration.

Answer: B,C,E

Explanation:

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IPsec VTIs have many benefits:

QUESTION NO: 92

In Cisco IOS 15.0.1M code for the router platform, which new feature has been added to the zone-based policy firewall?

- A. removal of support for port-to-application matching
- B. ability to configure policies for traffic that is traveling between interfaces in the same security zone
- C. intrazone traffic is not freely permitted by default now
- D. NBAR is not compatible with transparent firewall

Answer: B

Explanation:

Page: 309, CCNP Security SECURE 642-637 Official Cert Guide

With the release of IOS 15.0.1M, it is also possible to control the traffic within the same zone; this is referred to as intrazone. This is configured by creating a zone pair with the same two zone names as both source and destination.

QUESTION NO: 93

When configuring NAT, which three protocols that are shown may have limitations or complications when using NAT? (Choose three.)

- A. Kerberos
- B. HTTPS
- C. NTP
- D. SIP
- E. FTP
- F. SQL

Answer: A,D,E

Explanation:

As with any technology, the use of NAT can introduce problems because some technologies do not support the use of NAT. These limitations include:

QUESTION NO: 94

Which two answers are potential results of an attacker that is performing a DHCP server spoofing attack? (Choose two.)

- A. ability to selectively change DHCP options fields of the current DHCP server, such as the giaddr field.
- B. DoS

- C. excessive number of DHCP discovery requests
- D. ARP cache poisoning on the router
- E. client unable to access network resources

Answer: B,E

Explanation:

DHCP Server Spoofing

With DHCP server spoofing, the attacker can set up a rogue DHCP server and respond to DHCP requests from clients on the network. This type of attack can often be grouped with a DHCP starvation attack because the victim server will not have any new IP addresses to give out, which raises the chance of new clients using the rogue DHCP server. This information, which is given out by the rogue DHCP server, could send all the traffic through a rogue gateway, which can then capture the traffic for further analysis.

QUESTION NO: 95

Cisco IOS Software displays the following message: DHCP_SNOOPING_5-DHCP_SNOOPING_MATCH_MAC_FAIL. What does this message indicate?

- A. The message indicates that an attacker is pretending to be a DHCP server on an untrusted port.
- B. The source MAC address in the Ethernet header does not match the address in the "chaddr" field of the DHCP request message.
- C. The message indicates that the DHCP snooping has dropped a DHCP message that claimed an existing, legitimate host is present on an unexpected interface.
- D. A Layer 2 port security MAC address violation has occurred on an interface that is set up for untrusted DHCP snooping.

Answer: B

Explanation:

Actual Log from Switch configured for DHCP spoofing

```
007850: Nov 26 09:02:55.484 CET: %DHCP_SNOOPING-5-  
DHCP_SNOOPING_MATCH_MAC_FAIL:
```

DHCP_SNOOPING drop message because the chaddr doesn't match source mac, message type: DHCPRELEASE, chaddr: 0016.4487.6527, MAC sa: 0017.422e.d204

The switch logging message basically says that the MAC address of the client contained in the chaddr (client hardware address) field in the DHCP message does not match the source MAC address of the frame in which the DHCP message is encapsulated. In other words, the interface for which the DHCP message was created does not match the interface through which the message was actually transmitted.

<https://supportforums.cisco.com/thread/344460>

QUESTION NO: 96

Refer to the exhibit.

```
switch(config)#Interface range G1/0/1 - 24
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 30
Switch(config-if)#authentication port-control auto
Switch(config-if)#dot1x timeout quiet-period 5
Switch(config-if)#dot1x timeout tx-period 10
Switch(config-if)#authentication periodic
Switch(config-if)#authentication timer reauthenticate 600
Switch(config-if)#authentication event fail retry 0 action authorize vlan 40
Switch(config-if)#authentication even no-response action authorize vlan 40
```

Based on the partial configuration that is provided, if a non-802.1X client connects to a port on this switch, which VLAN will it be assigned to, and how long will it take for the port to time out and transition to the guest VLAN? (Choose all that apply.)

- A. The switch is configured for the default 802.1X timeout period of 90 seconds.
- B. The 802.1X authentication process will time out in 10 seconds and immediately change the port to the guest VLAN.
- C. The 802.1X authentication process will time out, and the switch will roll over the port to the guest VLAN in 15 seconds.
- D. The non-802.1X client and phones will all be assigned to VLAN 30.
- E. The non-802.1X client will be assigned to VLAN 40.
- F. The non-802.1X client will be assigned to VLAN 10.

Answer: C,E

Explanation:

The authenticator expects to receive the EAP-Response/Identity frame as a response to its EAP-Request/Identity frame. If it has not received this frame within the default retransmission time, it will resend the Request frame. The default retransmission timer is 30 seconds.

You can adjust this time to increase response times, which will allow a faster 802.1X

authentication process. The retransmission timer is changed with the dot1x timeout txperiod interface command.

If the switch fails to authenticate a client, such as the user entering a bad password, the switch waits a period of time before trying again. The default value for this quiet timer is 60 seconds. You can lower this value, thus giving the client a faster response time with the dot1x timeout quiet-period seconds interface configuration command.

QUESTION NO: 97

When 802.1X is implemented, how do the authenticator and authentication server communicate?

- A. RADIUS
- B. TACACS+
- C. MAB
- D. EAPOL

Answer: A

Explanation:

Page: 119

Note: EAPOL is used between the supplicant and the authenticator, while RADIUS is used between the authenticator and the authentication server.

QUESTION NO: 98

Refer to the exhibit.

```
ida-client server url https://www.cisco.com/cgi-bin/front.x/ida/locator/locator.pl
!
ip ips auto-update
  cisco
  occur-at weekly 0-6 1 0
  username CISCO123 password CISCOPASS
```

What can be determined about IPS updates from the configuration shown?

- A. Updates will be stored on the ida-client server.
- B. Updates will be stored in the directory labeled "cisco."
- C. Updates will be retrieved from an external source every day of the week.
- D. Updates will occur once per week on Sundays between midnight and 6 a.m. (0000 and 0600).

Answer: C

Explanation:

Task 2: Configure Automatic Signature Updates

The second task illustrates how to configure the router to attempt to retrieve automatic signature updates from Cisco.com or a local server.

To do this, first configure the update URL using the `ida-client server url` command. Use the `https://www.cisco.com/cgi-bin/front.x/ids/locator/locator.pl` URL. Next, create an auto-update profile using the `ip ips auto-update` command. Use the `cisco` command inside the profile to designate obtaining updates from Cisco.com. To control when the update attempts occur, use the `occur-at` command. Example 13-9 illustrates the setup of the configuration to retrieve automatic updates from the Cisco.com repository as well as to provide the Cisco.com credentials that will be used for authentication through using the `username` command.

Example 13-10 illustrates the setup of the configuration to retrieve automatic updates from a local staging server.

The following specifics are used in the example:

QUESTION NO: 99

Refer to the exhibit.

```
crypto isakmp policy 101
  encr 3des
  hash sha-1
  authentication rsa-sig
  group 2
  lifetime 864000
```

Which of these is correct based on the partial configuration shown?

- A. The policy is configured to use an authentication key of "rsa-sig."
- B. The policy is configured to use hashing group sha-1.
- C. The policy is configured to use triple DES IPsec encryption.
- D. The policy is configured to use digital certificates.
- E. The policy is configured to use access list 101 to identify the IKE-protected traffic.

Answer: D

Explanation:

QUESTION NO: 100

When uploading an IPS signature package to a Cisco router, what is required for the upload to self-extract the files?

- A. the idconf on the end of the copy command
- B. a public key on the Cisco router
- C. IPS must be disabled on the upload interface
- D. HTTP Secured server must be enabled

Answer: A

Explanation:

First, the signature package must be downloaded from Cisco.com. Go to the download section of Cisco.com and navigate to Products > Security > Integrated Router/Switch Security > Integrated Threat Control > Cisco IOS Intrusion Prevention System Feature Software > IOS IPS Signature Data File. Download the latest package, which should have a filename in the format IOS-Sxxx-CLI.pkg. Put the file on the server from which you will transfer it to the router. Use the copy command to transfer the file to the router's idconf alias. This causes the router to download and unpack the contents of the file (XML files)

QUESTION NO: 101

To prevent a spanning-tree attack, which command should be configured on a distribution switch port that is connected to an access switch?

- A. spanning-tree portfast bpduguard default
- B. spanning-tree backbone fast
- C. spannning-tree bpduguard enable
- D. spanning-tree guard root

Answer: D

Explanation:

To mitigate STP manipulation, two different features can be used. The Root Guard feature is configured on a switchport that should never become a root port, or in other words, the port that forwards traffic going toward the root bridge. A good example of this would be a connection between a distribution layer switch and an access layer switch. In this scenario, the port on the distribution switch going toward the access layer should never become a root port because the access layer switch should never become the root switch. If the switchport does receive a superior BPDU, the port will go into root-inconsistent state, indicating that another switch is attempting to become the root switch.

Enables the Root Guard feature on a switchport

Switch(config-if)# spanning-tree guard root

QUESTION NO: 102

In a GETVPN solution, which two ways can the key server distribute the new keys to the group members during the rekey process? (Choose two.)

- A. multicast UDP transmission
- B. multicast TCP transmission
- C. unicast UDP transmission
- D. unicast TCP transmission

Answer: A,C

Explanation:

Rekeying Methods

GET VPNs use rekey messages to refresh their IPsec SAs (session keys) outside of IKE sessions. When the group IPsec SAs are about to expire, one single rekey message for a particular group is generated on the key server. Distribution of the rekey message does not require that new IKE sessions be created. GET supports rekeying for Unicast and multicast.

QUESTION NO: 103

You are a network administrator and are moving a web server from inside the company network to a DMZ segment that is located on a Cisco router. The web server was located at IP address 172.16.10.50 on the inside and changed to the IP address 172.20.10.5 on the DMZ. Additionally,

you are moving the web port to 8080 but do not want your inside users to be affected. Which NAT statement should you configure on your router to support the change?

- A. hostname(config)# ip nat inside source static 172.16.10.50 172.20.10.5
- B. hostname(config)# ip nat inside source static tcp 172.16.10.50 80 172.20.10.5 8080
- C. hostname(config)# ip nat outside source static tcp 172.16.10.50 80 172.20.10.5 8080
- D. hostname(config)# ip nat static outside source tcp 172.20.10.5 80 172.16.10.50 8080
- E. hostname(config)# ip nat static inside source udp 172.20.10.50 172.20.10.5

Answer: B

Explanation:

QUESTION NO: 104

When configuring NAT, and your solution requires the ability to see the inside local and outside global address entries and any TCP or UDP port in the show ip nat command output, how should NAT be configured on the router?

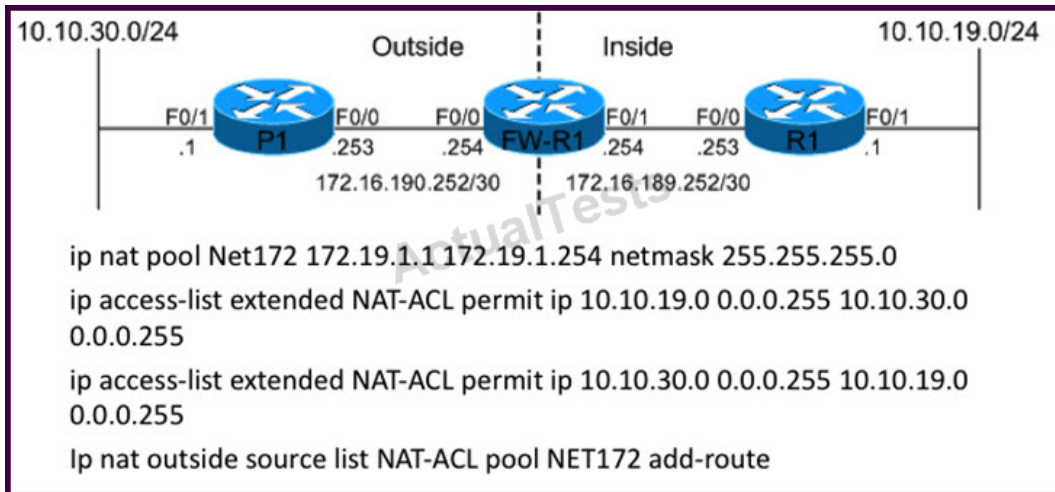
- A. use the overload option on the end of your static NAT statement
- B. include both static and dynamic NAT configuration on the router
- C. tie the ip nat inside command to a dynamic NAT pool
- D. attach a route-map to the ip nat inside command
- E. configure the ip nat inside command to an extended ACL

Answer: D

Explanation:

QUESTION NO: 105

Refer to the exhibit.



You are working for a corporation that has connected its network to a partner network. Based on this partial configuration that is supplied in the exhibit, which two things happen to traffic that is inbound from the partner network (outside is 10.10.30.0/24) and the return traffic from the inside as it travels through this router? (Choose two.)

- A. The source address of the IP packets that are traveling from the 10.10.30.0/24 network to 10.10.19.0/24 are translated to 172.19.1.0/24.
- B. The destination address of IP packets that are traveling from 10.10.19.0/24 to any IP network is translated to 172.19.1.0/24.
- C. IP traffic that is flowing from 10.10.19.0/24 to 10.10.30.0/24 has the source address translated to 172.19.1.0/24.
- D. The destination address of IP packets that are traveling from 10.10.19.0/24 to 10.10.30.0/24 are translated to 172.19.1.0/24.
- E. The destination address of IP packets that are traveling from 10.10.30.0/24 to 10.10.19.0/24 are translated to 172.19.1.0/24.

Answer: A,D

Explanation:

QUESTION NO: 106

You are a network administrator that is deploying a Cisco router that needs to support both PAT and site-to-site VPN on one public IP address. In order to make both work simultaneously, how should the NAT configuration be set up?

- A. The VPN configuration should be set up with a static NAT configuration.
- B. Because PAT does support AH, the VPN tunnel must not be configured with Encapsulating Security Payload (ESP).
- C. An ACL should be attached to the nat command to permit the NAT traffic and deny the VPN traffic.

- D. The nat configuration command needs to include a range of IP addresses with the overload word on the end.
- E. A route-map should be used with the nat command to support the use of AH and ESP.
- F. The ip nat inside command needs to exclude the VPN source address in the NAT pool.

Answer: C

Explanation:

QUESTION NO: 107

Refer to the exhibit.

```
switch(config)#interface GigabitEthernet1/0/10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport voice vlan 40
Switch(config-if)#authentication host-mode multi-domain
Switch(config-if)#authentication open
Switch(config-if)#authentication order mab dot1x
Switch(config-if)#authentication periodic
Switch(config-if)#authentication timer reauthenticate server
Switch(config-if)#mab
Switch(config-if)#dot1x pae authenticator
```

Based on the configuration that is shown in the exhibit, select the three answers that apply. (Choose three.)

- A. The configuration supports multidomain authentication, which allows one MAC address on the voice VLAN and one on the data VLAN.
- B. Traffic will not flow for either the phone or the host computer until one device completes the 802.1X authentication process.
- C. Registration and DHCP traffic will flow on either the data or voice VLAN before authentication.
- D. The port will only require the 802.1X supplicant to authenticate one time.
- E. MAC Authentication Bypass will be attempted only after 802.1X authentication times out.
- F. Non-802.1X devices are supported on this port by setting up the host for MAC address authentication in the endpoint database.

Answer: A,C,F

Explanation:

QUESTION NO: 108

You are finding that the 802.1X-configured ports are going into the error-disable state. Which command will show you the reason why the port is in the error-disable state, and which command will automatically be re-enabled after a specific amount of time? (Choose two.)

- A. show error-disable status
- B. show error-disable recovery
- C. show error-disable flap-status
- D. error-disable recovery cause security-violation
- E. error-disable recovery cause dot1x
- F. error-disable recovery cause l2ptguard

Answer: B,D

Explanation:

QUESTION NO: 109

Your company has a requirement that if security is compromised on phase 1 of a Diffie-Hellman key exchange that a secondary option will strengthen the security on the IPsec tunnel. What should you implement to ensure a higher degree of key material security?

- A. Diffie-Hellman Phase II ESP
- B. PFS Group 5
- C. Transform-set SHA-256
- D. XAUTH with AAA authentication
- E. Diffie-Hellman Group 5 Phase I

Answer: B

Explanation:

IPsec Phases

IPsec has two phases:

IPsec session keys are derived from the initial keying material that was obtained during the Phase 1 Diffie-Hellman key exchange. The IPsec session keys can be optionally created using new, independent Diffie-Hellman key exchanges by enabling the Perfect Forward Secrecy (PFS) option. This Phase 2 exchange is called the IKE Quick Mode. IKE Quick Mode is one of two modes of IKE Phase 2, with the other being the Group Domain of Interpretation (GDOI) Mode used by GET VPN.

QUESTION NO: 110

Which solution on a Cisco router requires the loading of a protocol header definition file (PHDF)?

- A. Reflexive access control lists
- B. NetFlow
- C. Flexible Packet Matching
- D. Control Plane Policing

Answer: C

Explanation:

FPM is implemented using a filtering policy that is divided into four tasks:

QUESTION NO: 111

You are troubleshooting a problem for which end users are reporting connectivity issues. Your network has been configured with Layer 2 protection controls. You have determined that the DHCP snooping database is correct and that proper static addressing maps have been configured. Which of these should be your next step in troubleshooting this problem?

- A. Generate a proxy ARP request and verify that the DHCP database has been updated as expected.
- B. Temporarily disable DHCP snooping and test connectivity again.
- C. Clear the ARP tables and have end users release and renew their DHCP-learned addressing.
- D. Use a protocol analyzer to determine if there are malformed DHCP or ARP packets.

Answer: D

Explanation:

QUESTION NO: 112

You are troubleshooting a reported connectivity issue from a remote office whose users are accessing corporate headquarters via an IPsec VPN connection. You issued a show crypto isakmp sa command on the headend router, and the state has MM_NO_STATE. Which debug command should you enter next, and which part of the VPN tunnel establishment process is

failing? (Choose two.)

- A. ISAKMP Phase II
- B. ISAKMP Phase I
- C. debug crypto isakmp sa
- D. debug crypto isakmp
- E. debug crypto ipsec

Answer: B,D

Explanation:

Troubleshooting Flow

Follow these steps to proceed through the recommended flow for troubleshooting IKE peering:

Step 1. Verify peer reachability using the ping and traceroute commands with the tunnel source and destination IP addresses on both peers. If connectivity is verified, proceed to Step 2; otherwise, check the path between the two peers for routing or access (firewall or access list) issues.

Step 2. Verify the IKE policy on both peers using the show crypto isakmp policy command. Debug messages revealed by the debug crypto isakmp command will also point out IKE policy mismatches.

Step 3. Verify IKE peer authentication. The debug crypto isakmp command will display unsuccessful authentication.

Step 4. Upon successful completion of Steps 1–3, the IKE SA should be establishing. This can be verified with the show crypto isakmp sa command and looking for a state of QM_IDLE.

QUESTION NO: 113

You are installing a brand-new, site-to-site VPN tunnel and notice that it is not working correctly. When connecting to the corporate router and issuing a show crypto ipsec sa command, you notice that for this particular SA that packets are being encrypted but not decrypted. What are two potential reasons for this problem? (Choose two.)

- A. XAUTH needs to be enabled.
- B. Inbound and outbound IP 50 packets are being filtered at the remote site.
- C. The transform-set needs to be set to transport mode.
- D. The access-list attached to the crypto map at the remote site is incorrect.
- E. The remote site is failing Diffie-Hellman Phase I negotiation.
- F. The NAT exception on the corporate side is filtering the return packets.

Answer: B,D

Explanation:

QUESTION NO: 114

Which two of these are features of control plane security on a Cisco ISR? (Choose two.)

- A. CoPP
- B. RBAC
- C. AAA
- D. CPPr
- E. uRPF
- F. FPM

Answer: A,D

Explanation:

QUESTION NO: 115

Which additional configuration steps are required for a zone-based policy firewall to operate in a VRF scenario?

- A. You must assign zone-based policy firewall bridge groups to work in the virtual environment.
- B. Separate zone-based policy firewall policies must be defined for each VRF environment.
- C. Separate zones must be defined for each virtual zone-based policy firewall instance.
- D. No special zone-based policy firewall configurations are needed.

Answer: D

Explanation:

Ensure that you utilized several security layers in your design to adequately protect the rest of your network from the guest VLAN. You might even consider putting them in a separate Virtual Routing and Forwarding (VRF) instance. VRFs are configurations on Cisco IOS Software routers and switches that can be used to provide traffic separation, making them a good solution to keep guest traffic segregated from your corporate traffic.

ZBPFW is also Virtual Routing and Forwarding (VRF) aware and can be used between different VRFs. Interfaces that are configured in different VRFs should not be configured in the same zone, and thus all interfaces that are in a zone must be configured within the same VRF. If there is a common interface or interfaces that are used by multiple VRFs, a common zone should be created and individually paired with each zone (and thus with each VRF).

QUESTION NO: 116

You are troubleshooting an IPsec VPN problem. During debugging of IPsec operations, you see the message "attributes not acceptable" on the IKE responder after issuing the debug crypto isakmp command. Which step should you take next?

- A. verify matching ISAKMP policies on each peer
- B. verify that an IKE security association has been established between peers
- C. verify that IPsec transform sets match on each peer
- D. verify if default IPsec attributes are in place on each peer

Answer: C

Explanation:

The show crypto isakmp policy command can be executed on both peers to compare IKE parameters and ensure that they match. The debug crypto isakmp debugging command will display debugging messages during IKE negotiation and session establishment. These debugging commands should be executed and analyzed on both peers.

QUESTION NO: 117

Which state is a Cisco IOS IPS signature in if it does not take an appropriate associated action even if it has been successfully compiled?

- A. retired
- B. disabled
- C. unsupported
- D. inactive

Answer: B

Explanation:

QUESTION NO: 118

Which CLI command would you use to verify installed SSL VPN licensing on a Cisco 1900, 2900, or 3900 Series ISR?

- A. show crypto ssl license
- B. show crypto webvpn details
- C. show webvpn license
- D. show webvpn ssl license count all
- E. show webvpn gateway

Answer: C

Explanation:

http://www.cisco.com/en/US/docs/ios-xml/ios/sec_conn_sslvpn/configuration/15-1mt/sec-conn-sslvpn-15-1mtbook.pdf

You can use the show webvpn license command to display the available count and the current usage. To display the current license type and time period left in case of a nonpermanent license, use the show license command. To get information related to license operations, events, and errors, use the debug webvpn license command.

For migrating from any Cisco IOS 12.4T release to Cisco IOS 15.x release, use the license migration tool at

<https://tools.cisco.com/SWIFT/Licensing/LicenseAdminServlet/migrateLicense>.

New Cisco IOS SSL VPN licenses that are generated are cumulative. Therefore the old licenses become inactive when a new license is applied. For example, when you are upgrading your license from 10 counts to 20 counts (an increase of 10 counts on the current 10 counts), Cisco provides a single 20 count license.

The old license for 10 counts is not required when a permanent license for a higher count is available.

However, the old license will exist in an inactive state as there is no reliable method to clear the old license.

In Cisco IOS Release 15.1(4)M1 and later releases, a Crypto Export Restrictions Manager (CERM) license is reserved only after the user logs in. If you have an Integrated Services Router Generation 2 (ISR G2) router with a CERM license, you must upgrade to Cisco IOS Release 15.1(4)M1 or later releases. Before Cisco IOS Release 15.1(4)M1, a CERM license is reserved for every SSL or Transport Layer Security (TLS) session.

QUESTION NO: 119

Which statement is correct regarding GRE tunnel endpoints when you are configuring GRE over IPsec?

- A. The tunnel interfaces of both endpoints must be in the same IP subnet.
- B. A mirror image of the IPsec crypto ACL needs to be configured to permit the interesting end-user traffic between the GRE endpoints.
- C. The tunnel interfaces of both endpoints should be configured to use the outside IP address of

the router as the unnumbered IP address.

D. For high availability, the GRE tunnel interface should be configured with a primary and a backup tunnel destination IP address.

Answer: A

Explanation:

QUESTION NO: 120

Refer to the exhibit.

```
crypto pki trustpoint MY-TRUSTPOINT
rsa-keypair CS-KEYS
!
ip http server
!
crypto pki server MY-TRUSTPOINT
issuer-name CN=MY-CS,OU=VPN,O=Cisco,C=US
database url flash:/my-ca
database level complete
hash sha1
lifetime certificate 730
lifetime ca-certificate 3650
no grant auto
lifetime crl 5
```

Which of these is correct regarding the configuration parameters shown?

- A.** Complete certificates will be written to and stored in NVRAM.
- B.** The RSA key pair is valid for five hours before being revoked.
- C.** The router is configured as a certificate server.
- D.** Certificate lifetimes are mismatched and will cause intermittent connectivity errors.
- E.** The router has enrolled to the MY-TRUSTPOINT PKI server, which is an external CA server.

Answer: C

Explanation:

QUESTION NO: 121

Refer to the exhibit.

```
Router1#show ip interface brief | include Virtual-Access
Virtual-Access1      unassigned      YES unset  down  down
Virtual-Access2      192.168.1.1    YES unset  up    up
```

When you are using dynamic IPsec VTI tunnels, what can you determine about virtual-access interfaces from the output shown?

- A. The Virtual-Access1 interface currently does not have an IPsec peer connection established.
- B. The Virtual-Access2 interface does not yet have an IPsec peer defined.
- C. The Virtual-Access1 interface is in the down/down state, because the virtual tunnel source physical interface is down.
- D. The Virtual-Access1 interface, which is used internally by the Cisco IOS software, is always down.

Answer: A

Explanation: "A special Virtual-Access1 interface is used internally by Cisco IOS Software and is always present in the output of this command." but not always DOWN !!!

as follows from:

http://www.cisco.com/en/US/prod/collateral/iosswrel/ps6537/ps6586/ps6635/prod_white_paper0900aecd803645b5.pdf

"...When the Easy VPN negotiation is successful, the line protocol state of the virtual access interface gets changed to up. When the Easy VPN tunnel goes down because the security association expires or is deleted, the line protocol state of the virtual access interface changes to down..."

QUESTION NO: 122

Refer to the exhibit.

```
!  
crypto gdoi group CLASSROOM  
identity number 12345  
!  
crypto map GETVPN_MAP 10 gdoi  
set group CLASSROOM  
!  
interface Serial0/0/0  
ip address 192.168.1.2 255.255.255.252  
crypto map GETVPN_MAP  
!
```

Based on the partial configuration shown, which additional configuration parameter is needed under the GET VPN group member GDOI configuration?

- A. key server IP address
- B. local priority
- C. mapping of the IPsec profile to the IPsec SA
- D. mapping of the IPsec transform set to the GDOI group

Answer: A

Explanation:

QUESTION NO: 123 CORRECT TEXT

Scenario

You have been given the task of performing initial zone-based policy firewall configurations. You will need to create zones, assign the zones to specific interfaces, and create zone pairs to allow for traffic flow between interfaces. You will also need to define a zone-based policy firewall and assign the policy to the zone pair. To access the router console ports, refer to the exhibit, click the router for access, and perform the following tasks.

Note that when performing the configuration, you should use the exact names highlighted in bold below:

- Globally create zones and label them with the following names:
 - OUTSIDE
 - INSIDE
- Assign interfaces to zones as indicated in the exhibit
- Create a zone pair for traffic flowing from the inside to outside zones named **IN-TO-OUT**
- Define a zone-based firewall policy named **IN-TO-OUT-POLICY**
 - Use the 'match protocol' classification option to statfuly inspect HTTP traffic and drop all other traffic
 - Use a class-map named **HTTP_POLICY**
- Apply zone-based firewall policy **IN-TO-OUT-POLICY** to the zone pair

Topology

The diagram shows a central router with two interfaces. On the left, interface Fa0/0/0 is connected to a red box labeled 'OUTSIDE zone'. On the right, interface Fa0/0/1 is connected to a green box labeled 'INSIDE zone'.

```

Router(config)# zone security INSIDE
Router(config-sec-zone)#exit
Router(config)# zone security OUTSIDE
Router(config-sec-zone)#exit
Router(config)# interface fa0/0/1
Router(config-if)# no shutdown
Router(config-if)# zone-member security INSIDE
Router(config-if)# exit
Router(config)# interface fa0/0/0
Router(config-if)# no shutdown
Router(config-if)# zone-member security OUTSIDE
Router(config-if)# exit
Router(config)# class-map type inspect match-any HTTP_POLICY
Router(config-cmap)# match protocol http
Router(config-cmap)#exit
Router(config)# policy-map type inspect IN-TO-OUT-POLICY
Router(config-pmap)# class type inspect HTTP_POLICY
Router(config-pmap-c)# inspect
Router(config-pmap-c)# exit
Router(config)# zone-pair security IN-TO-OUT source INSIDE destination OUTSIDE
Router(config-sec-zone-pair)# service-policy type inspect IN-TO-OUT-POLICY
Router(config-sec-zone-pair)# end
Router(config)# copy running-config startup-config
  
```

Explanation:

1: we divide the network into 2 zones: INSIDE and OUTSIDE

- 2: apply the interfaces to the appropriate Zone Members INSIDE | OUTSIDE
- 3: create a class-map with defined name HTTP_POLICY > match HTTP protocol
- 4: create a policy-map name IN-TO-OUT-POLICY: - define the class-map and apply action > inspect
- 5: create a zone-pair > specify direction with source and destination
- 6: apply policy to the zone-pair - policy created in step 4
- 7: std: copy run start

QUESTION NO: 124 DRAG DROP

Associate the correct DMVPN parameter information on the left with the associated description on the right

NHRP Authentication string	When this expires, the network ID is no longer valid
NHRP Network ID	This needs to be same for all mGRE tunnels on the network
Tunnel Key	This is used for NBMA network
NHRP Hold Time	This is used for DMVPN tunnel hubs and spokes to authenticate themselves.
NHRP NHS	
NHRP Registration	

Answer:

Associate the correct DMVPN parameter information on the left with the associated description on the right

NHRP Authentication string	NHRP Hold Time
NHRP Network ID	Tunnel Key
Tunnel Key	NHRP NHS
NHRP Hold Time	NHRP Registration
NHRP NHS	
NHRP Registration	

Explanation:

Associate the correct DMVPN parameter information on the left with the associated description on the right

NHRP Authentication string

NHRP Network ID

NHRP Hold Time

Tunnel Key

NHRP NHS

NHRP Registration

untitled

http://www.cisco.com/en/US/docs/ios/12_4/ip_addr/configuration/guide/hadnhrp_ps6350_TSD_Products_Configuration_Guide_Chapter.html and <http://blog.ine.com/2008/08/02/dmvpn-explained/> shows this also is this:

6) NHRP Registration is used for ease of support/creation of dynamic tunnels but is not the same as 1) NHRP Authentication string. Authentication is recommended to help keep multiple NHRP domains separate from each other.

2) NHRP network ID is used to differentiate multiple NHRP domains So

a – 4

b – 3 Prior to Cisco IOS Release 12.3(11)T, all mGRE interfaces required the configuration of a tunnel ID key.

Multipoint tunnels require that you configure a tunnel key. Otherwise, unexpected GRE traffic could easily be received by the tunnel interface. However, for simplicity, it is recommended that the tunnel key correspond to the NHRP network identifier.

c – 5

d – 6

Original dump had 4,1,5,6 not 4,3,5,6 and this explanation

NHRP Hold Time – When this expires, the network ID is no longer valid

NHRP Authentication string - This needs to be the same for all mGRE tunnels on the network

NHRP NHS – This is used for NBMA network

NHRP Registration – This is used for DMVPN tunnel hubs and spokes to authenticate themselves

To make registration possible, you configure each NHC (Client/spoke) with the IP address of at least one NHS (server/hub). In turn, NHS acts as a database agent, storing all registered mappings, and replying to NHC queries.

The NHS will keep the registration request cached for the duration of the hold-time, and then, if no registration update is received, will time it out. One can adopt NHRP to work with “simulated NBMA” networks, such as mGRE tunnels.

The commands `ip nhrp network-id` and `ip nhrp authentication [Key]` identify and authenticate the logical NHRP network. The [ID] and the [Key] must match on all routers sharing the same GRE tunnel. It is possible to split an NBMA medium into multiple NHRP networks

The spokes use the NHS to register their logical IP to NBMA IP associations and send NHRP resolution Requests

QUESTION NO: 125 DRAG DROP

Drag the box on the left over to the box on the right that can cause a port to go into the error-disable state.

BPDU guard violation	Target
Inline power disabled, but device requires power	Target
Speed Mismatch	Target
DHCP snooping rate-limit reached	
Port channel misconfiguration	

Answer:

Drag the box on the left over to the box on the right that can cause a port to go into the error-disable state.

BPDU guard violation	BPDU guard violation
Inline power disabled, but device requires power	DHCP snooping rate-limit reached
Speed Mismatch	Port channel misconfiguration
DHCP snooping rate-limit reached	
Port channel misconfiguration	

Explanation:

BPDU guard violation

DHCP snooping rate-limit reached

Port channel misconfiguration

There are various reasons for the interface to go into errdisable. The reason can be:

Reference:

http://www.cisco.com/en/US/tech/tk389/tk621/technologies_tech_note09186a00806cd87b.shtml

QUESTION NO: 126

When you are configuring a hub-and-spoke DMVPN network, which tunnel mode should you use for the spoke router configuration?

- A. GRE multipoint
- B. Classis point-to-point GRE
- C. IPsec multipoint
- D. Nonbroadcast multiaccess

Answer: B

Explanation:

http://www.cisco.com/en/US/docs/solutions/Enterprise/WAN_and_MAN/DMVPN_2_Phase2.html

The hub-and-spoke deployment model is the most common deployment model. This model is the most scalable, and predominately mimics traditional Layer 2 leased line, Frame Relay, or ATM hub-and-spoke networks. The headend is configured with a multipoint GRE (mGRE) interface, and the branch with a point-to-point (p2p) GRE interface.

QUESTION NO: 127 DRAG DROP

Match the items on the left with their descriptions on the right.

True Positives

The IPS or IDS sensor acted as a consequence of non-malicious activity. This represents an error, generally caused by signatures that are too relaxed

True Negatives

The IPS or IDS sensor did not take action, even through there was malicious activity. This represents an error.

False Positives

The IPS or IDS sensor acted as a consequence of malicious activity. This represents normal and optimal operation.

False Negatives

The IPS or IDS sensor did not take action, because there was no malicious activity. This represents normal and optimal operation

Answer:

Match the items on the left with their descriptions on the right.

True Positives

False Positives

True Negatives

False Negatives

False Positives

True Positives

False Negatives

True Negatives

Explanation:

Match the items on the left with their descriptions on the right.

False Positives

False Negatives

True Positives

True Negatives

untitled

True Positives – The IPS or IDS sensor acted as a consequence of malicious activity. The represents normal and optimal operation.

True Negative – The IPS or IDS sensor did not take action, even though there was malicious activity. This represents an error.

False Positive – The IPS or IDS sensor acted as a consequence of non-malicious activity. This represents an error, generally caused by signatures that are too relaxed.

False Negative – The IPS or IDS sensor did not take action, because there was no malicious activity. This represents normal and optimal operation.

QUESTION NO: 128 DRAG DROP

Match the descriptions on the left with 802.1X port states on the right.

This is the default port state when 802.1X is not globally enabled

In this state, the port monitors all traffic, excluding any attempts to authenticate

In this state, the port ignores all traffic

In this mode, the port allows only EAPOL, Cisco Discovery Protocol, and STP traffic at first

In this mode, the port ignores all Cisco Discovery Protocol and STP and accepts from 802.1X supplicants

Auto

Force-Authorized

Force-Unauthorized

Answer:

Match the descriptions on the left with 802.1X port states on the right.

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Explanation:

Match the descriptions on the left with 802.1X port states on the right.

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In this state, the port monitors all traffic, excluding any attempts to authenticate

In this mode, the port ignores all Cisco Discovery Protocol and STP and accepts from 802.1X supplicants

untitled

http://www.cisco.com/en/US/docs/switches/lan/catalyst2950/software/release/12.1_9_ea1/configuration/guide/Sw8021x.html

- force-authorized—disables 802.1X and causes the port to transition to the authorized state without any authentication exchange required. The port transmits and receives normal traffic

without 802.1X-based authentication of the client. This is the default setting.

- **force-unauthorized**—causes the port to remain in the unauthorized state, ignoring all attempts by the client to authenticate. The switch cannot provide authentication services to the client through the interface.
- **auto**—enables 802.1X authentication and causes the port to begin in the unauthorized state, allowing only EAPOL frames to be sent and received through the port. The authentication process begins when the link state of the port transitions from down to up, or when an EAPOL-start frame is received. The switch requests the identity of the client and begins relaying authentication messages between the client and the authentication server. Each client attempting to access the network is uniquely identified by the switch by using the client's MAC address.

QUESTION NO: 129 DRAG DROP

Match the 802.1X solution on the left with the appropriate descriptions on the right.

Guest VLAN	This method is used when clients do not support the 802.1X supplicant but need to be authenticated to an 802.1X network
Restricted VLAN	This solution is used when users fail authentication and have an 802.1X-compliant device
MAC Authentication Bypass	This method offers limited access for users without an 802.1X client. By default, it takes 90 seconds for the machine to get assigned to this specific VLAN
Web Authentication	Clients that use this method can be reauthenticated. If reauthentication fails, then the switch can assign the port to the guest VLAN if it is not configured

Answer:

Match the 802.1X solution on the left with the appropriate descriptions on the right.

Guest VLAN	This method is used when clients do not support the 802.1X supplicant but need to be authenticated to an 802.1X network
Restricted VLAN	This solution is used when users fail authentication and have an 802.1X-compliant device
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Web Authentication	Clients that use this method can be reauthenticated. If reauthentication fails, then the switch can assign the port to the guest VLAN if it is not configured

Explanation:

Match the 802.1X solution on the left with the appropriate descriptions on the right.

Web Authentication

Restricted VLAN

Guest VLAN

MAC Authentication Bypass

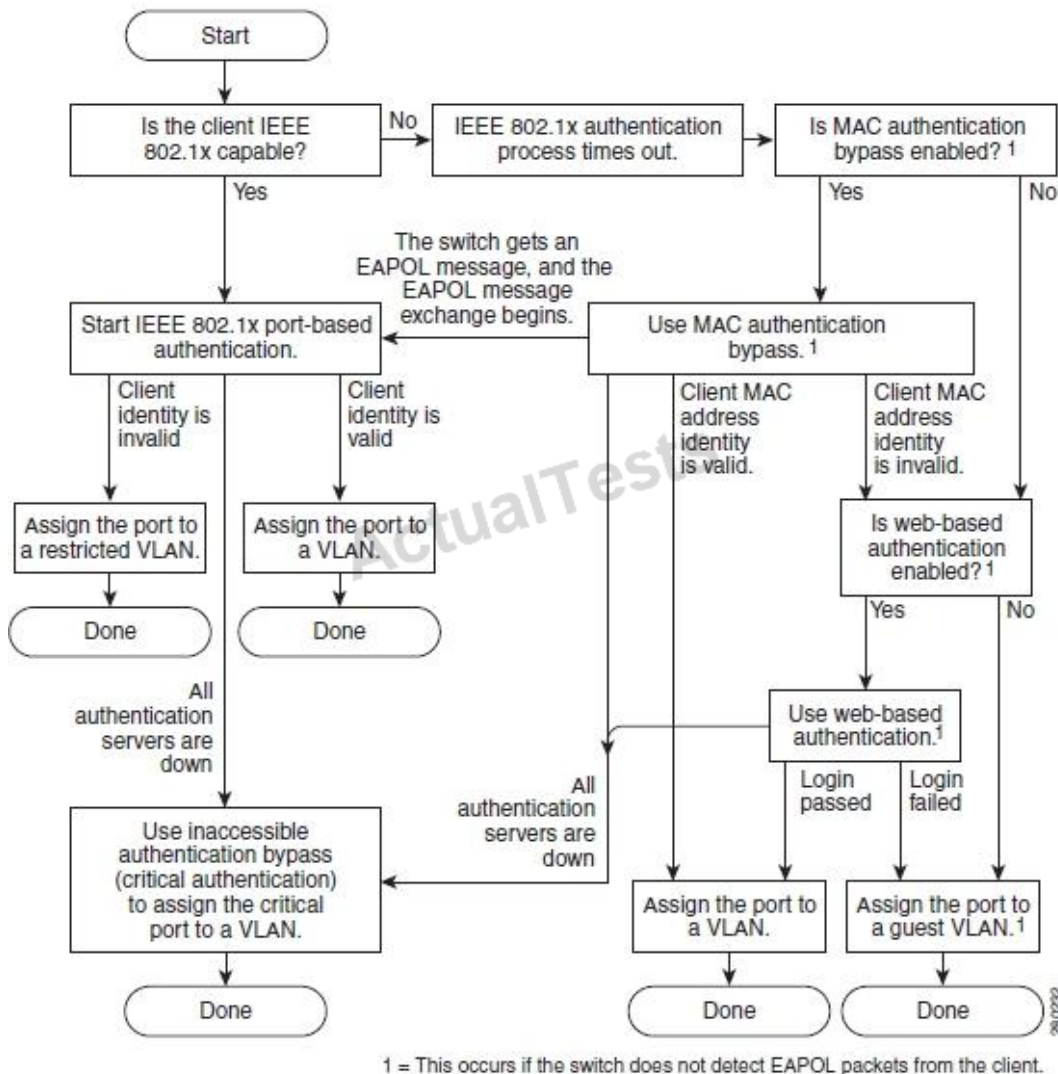
untitled

The default order for authentication methods is 802.1X, and then MAB, then web-based authentication.

If fallback authentication methods are not enabled or are not successful, and if a guest VLAN is configured, the switch assigns the client to a guest VLAN that provides limited services.

If the switch receives an invalid identity from an 802.1X-capable client and a restricted VLAN is specified, the switch can assign the client to a restricted VLAN that provides limited services.

If the RADIUS authentication server is unavailable (down) and inaccessible authentication bypass is enabled, the switch grants the client access to the network by putting the port in the critical-authentication state in the user-specified critical VLAN. Release 12.2(33)SXJ1 and later releases support configuration of critical voice and data VLANs.



untitled

Restrict

If the switch receives an invalid identity from an 802.1X-capable client and a restricted VLAN is specified, the switch can assign the client to a restricted VLAN that provides limited services.

Guest

If fallback authentication methods are not enabled or are not successful, and if a guest VLAN is configured, the switch assigns the client to a guest VLAN that provides limited services.

NOTE: You can configure a VLAN to be both the guest VLAN and the restricted VLAN if you want to provide the same services to both types of users.

If 802.1X authentication times out while waiting for an EAPOL message exchange, the switch can use a fallback authentication method, such as MAC authentication bypass (MAB) or web-based authentication (webauth), if either or both are enabled:

If MAC authentication bypass is enabled, the switch relays the client's MAC address to the AAA server for authorization. If the client's MAC address is valid, the authorization succeeds and the switch grants the client access to the network.

If web-based authentication is enabled, the switch sends an HTTP login page to the client. The

switch relays the client's username and password to the AAA server for authorization. If the login succeeds, the switch grants the client access to the network.

Reference:

<http://www.cisco.com/en/US/docs/switches/lan/catalyst6500/ios/12.2SX/configuration/guide/dot1x.html#wp1133480>

QUESTION NO: 130 DRAG DROP

Drag the items on the left to the boxes on the right that represents security features of a Cisco Catalyst Switch and Cisco IBR network device's control plane. Not all items will be used

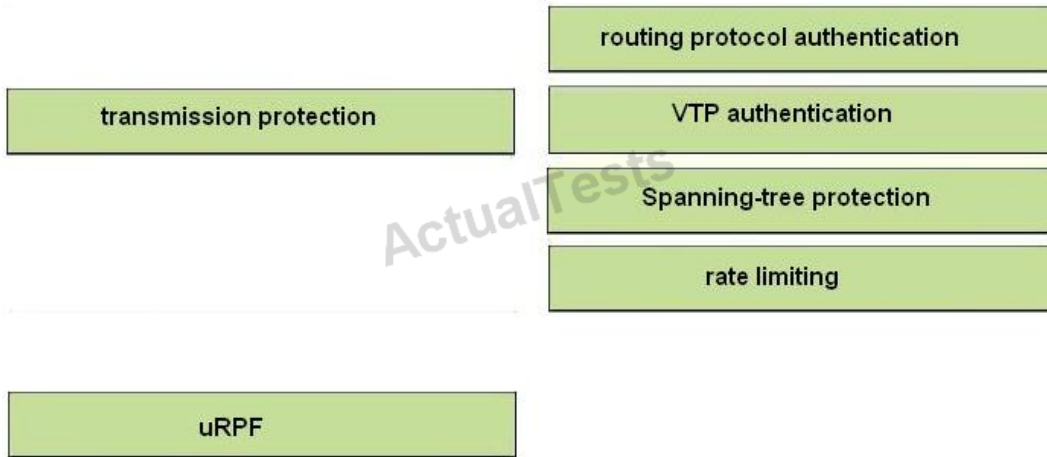
routing protocol authentication	
transmission protection	
rate limiting	
VTP authentication	
Spanning-tree protection	
uRPF	

Answer:

Drag the items on the left to the boxes on the right that represents security features of a Cisco Catalyst Switch and Cisco IBR network device's control plane. Not all items will be used

routing protocol authentication	routing protocol authentication
transmission protection	VTP authentication
rate limiting	Spanning-tree protection
VTP authentication	rate limiting
Spanning-tree protection	
uRPF	

Explanation:



untitled

http://www.cisco.com/web/about/security/intelligence/coppwp_gs.html

QUESTION NO: 131 DRAG DROP

Match the EAP type on the left with the associated description on the right. Not all EAP Types are used.

EAP-MD5	This Method requires that the channel be encrypted and integrity with TLS. Then a new EAP type is used for authenticating network access attempts by the client
EAP-TLS	This method encrypted EAP transactions within a TLS. The client uses protected access credentials (PAC) to encrypt EAP transactions
EAP-TTLS	This method provides mutual authentication and server through the use of a public key infrastructure (PKI). Both must have a certificate for this to work.
EAP-FAST	This method is a standard-based, nonproprietary type that all EAP deployments should support.
EAP-GTC	
PEAP	

Answer:

Match the EAP type on the left with the associated description on the right. Not all EAP Types are used.

EAP-MD5	T ir a EAP-TTLS
EAP-TLS	T Encrypt EAP transactions EAP-FAST
EAP-TTLS	T tl n EAP-TLS
EAP-FAST	T t EAP-MD5
EAP-GTC	
PEAP	

Explanation:

Box 1 – EAP-TTLS

Box 2 – EAP-FAST

Box 3 – EAP-TLS

Box 4 – EAP-MD5

EAP-MD5

EAP-MD5 is the only IETF Standards Track based EAP method - it is not recommended for use by Cisco

PEAP

The Protected Extensible Authentication Protocol, also known as Protected EAP or simply PEAP, is a protocol that encapsulates EAP within a potentially encrypted and authenticated Transport Layer Security (TLS) tunnel. PEAP uses the TLS channel to protect a second EAP exchange, called the "inner" EAP exchange. PEAP's major advantage is support from Microsoft

EAP-GTC

EAP-GTC, is an EAP method created by Cisco as an alternative to PEAPv0/EAP-MSCHAPv2. EAP-GTC carries a text challenge from the authentication server, and a reply generated by a security token.

EAP-FAST

EAP-FAST (Flexible Authentication via Secure Tunneling) is a protocol proposal by Cisco Systems as a replacement for LEAP

The protocol was designed to address the weaknesses of LEAP while preserving the "lightweight" implementation. Use of server certificates is optional in EAP-FAST. EAP-FAST uses a Protected Access Credential (PAC) to establish a TLS tunnel in which client credentials are verified. EAP-FAST has three phases.

EAP-TTLS

EAP-Tunneled Transport Layer Security (EAP-TTLS) is an EAP protocol that extends TLS. It was co-developed by Funk Software and Certicom. It is widely supported across platforms, although there is no native OS support for this EAP protocol in Microsoft Windows. The client can but does not have to be authenticated via a CA-signed PKI certificate to the server. This greatly simplifies the setup procedure as a certificate does not need to be installed on every client. TTLS implementations today support all methods defined by EAP, as well as several older methods (CHAP, PAP, MS-CHAP and MS-CHAPv2).

EAP-TLS

EAP-Transport Layer Security (EAP-TLS), defined in RFC 5216, is an IETF open standard, and is well-supported among wireless vendors. The security of the TLS protocol is strong, provided the user understands potential warnings about false credentials. It uses PKI to secure communication to a RADIUS authentication server or another type of authentication server. Unlike most HTTPS client implementations like major web browsers, most EAP-TLS implementations require client certificates, which some have identified as potentially dramatically reducing adoption of EAP-TLS. EAP-TLS is the original, standard wireless LAN EAP authentication protocol. Although it is rarely deployed, it is still considered one of the most secure EAP standards available and is universally supported by all manufacturers of wireless LAN hardware and software.

QUESTION NO: 132

Refer to the exhibit.

```
Interface FastEthernet0/0
Bridge-group 1
Zone member security OUT SIDE
!
Interface FastEthernet0/1
Bridge-group 1
Zone member security IN SIDE
```

What can be determined from the partial configuration shown?

- A. The zone-based policy firewall is operating in transparent mode.
- B. The zone-based policy firewall is providing for bridging of non-IP protocols.

- C. Since the interfaces are in the same bridge group, access policies are not required.
D. Traffic flow will be allowed to pass between the interfaces without being inspected.

Answer: A

Explanation:

QUESTION NO: 133 DRAG DROP

Drag the items on the left to the boxes on the right that define security of a network device's data plane.
Not all items will be used.

traffic filtering features	
routing protocol authentication	
transmission protection	
traffic conditioning features	
role-based access control	
protection against attacks on network endpoints	

Answer:

Drag the items on the left to the boxes on the right that define security of a network device's data plane.
Not all items will be used.

traffic filtering features	traffic filtering features
routing protocol authentication	transmission protection
transmission protection	traffic conditioning features
traffic conditioning features	protection against attacks on network endpoints
role-based access control	
protection against attacks on network endpoints	

Explanation:

Traffic filtering measures

Transmission protection

Traffic conditioning features

Protection against attacks on network endpoints

QUESTION NO: 134 DRAG DROP

What are the following items that are recommended in the DHCP snooping implementation guidelines?
Drag the recommended items on the left over to the right.

In a multiswitch environment, designate the interswitch links as trusted

Target

In a multiswitch environment, downstream ports to switches that do not support DHCP snooping should be trusted

Target

The preferred option to prevent DHCP starvation is port security rather than DHCP are limiting

Target

NTP should be configured on switches to ensure that correct handling of DHCP snooping database

ARP inspection rate limiting is the preferred way od handling DHCP starvation.

Answer:

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Drag the recommended items on the left over to the right.

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ARP inspection rate limiting is the preferred way od handling DHCP starvation.

NTP should be configured on switches to ensure that correct handling of DHCP snooping database

ARP inspection rate limiting is the preferred way od handling DHCP starvation.

Explanation:

In a multiswitch environment, designate the interswitch links as trusted

NTP should be configured on switches to ensure that coorrect handling of DHCP snooping database

ARP inspection rate limiting is the preferred way of handling DHCP starvation.

